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Design of an IoT sensor-based network architecture for waste collection in Bukavu city (DRC)

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Abstract

Bukavu (DRC) experiences significant waste management challenges, exacerbated by rapid urbanization. The city produces approximately 898 tons of waste daily. Yet only 7% is collected through an inefficient system, increasing sanitation and environmental risks. This study designs and validates a comprehensive IoT network architecture. The system enables real-time monitoring of waste bin fill levels and optimizes collection routes. The methodology involved a 12 week pilot deployment of seven prototypes across three neighborhood types: residential, dense, and peri-urban. The system uses ultrasonic sensors with a two point calibration method to accurately estimate bin fill levels. A set of seven 120 L smart bins was deployed. The results provide evidence for the system's robustness. Sensor energy autonomy remained above 90% after 11 months for most units, despite network challenges where 57% of sensors had Received-Signal-Strength-Indicator less than or equal to -107 decibel-milliwatt and 43% exhibited negative Signal-to-Noise Ratio values down to −10.25 decibel. It also enabled waste quantification, with an estimated 641 kg calculated from sensor fill levels over 14 days in one pilot area (seven households). The 12 week pilot across three neighborhood types validates the technical framework for a city-wide deployment. A theoretical economic analysis projects a 99% reduction in cost per ton, a 4 month payback period, and annual savings, enabling an increase in the collection rate from 7% to 30%. The architecture's scalability is shown to be technically feasible for city-wide deployment based on pilot data and network capacity modelling, and a preliminary security risk assessment is presented. All waste mass, volume, and route time values reported are estimates or calculations derived from sensor data, not direct measurements.

Keywords Internet of Things, Smart waste management, LoRaWAN, Ultrasonic sensors, Waste collection route optimization, Real-time monitoring

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1 Introduction

Population growth and rapid urbanization are driving the increase in global waste production [1]. According to a recent report by the United Nations [2], the global urban population will increase from 56% in 2021 to 68% in 2050. This could increase the volume of waste produced globally to 3.4 billion tons [3]. Waste management is therefore a major challenge worldwide and particularly in the Democratic Republic of the Congo (DRC).

Bukavu (eastern DRC, 26,667 inhabitants/km^2 [4]) produces 898 tons per day [5]. Production drivers include living standards, population growth, and rural exodus [6]. In Bukavu, two main types of waste are produced [6]: biodegradable (organic) and non-biodegradable (plastics, wood, glass, paper, textiles, and metal). Wood waste from processing is another type of waste produced in the city [7].

Bukavu's terrain complicates collection [4, 5, 8]. Three municipalities (Kadutu, Ibanda, and Bagira) host 23 collection organizations for 1,369,430 inhabitants [5, 9]. Collection occurs once weekly (7am-5pm), covering only 7 % (62.86 tons/day) [5].

This low collection rate stems from multiple factors [6]. These include a degraded road network, a lack of secondary roads, residents' living standards, and low public awareness of proper waste disposal.

Among the 39 total dump sites [10], most are located in inappropriate areas such as gutters, water collection points, and public squares. Environmentally, these sites lack mechanisms for waste sorting or recovery [10]; composting is the only technique currently used to reduce waste volume and recover materials. Unenforced dumping promotes microbes, malaria, and cholera [10].

This inefficiency degrades the urban environment and creates data gaps [11]. This study addresses the following research question: How can we design an economically viable and technologically robust IoT architecture for real-time waste monitoring and collection route optimization in Bukavu?

We hypothesize that combining LoRaWAN sensors with an open-source management platform will increase collection efficiency and reduce operational costs by at least 30%.

More specifically, this study makes three main contributions. First, it proposes an IoT architecture that incorporates sensors, communication modules, and a data management platform. This design accounts for the technical, economic and energy constraints of an urban African context, using Bukavu as a case study. Second, it selects and justifies appropriate technologies (ultrasonic sensors, LoRaWAN/MQTT protocols, and the OpenRemote IoT platform), considering Bukavu's limited infrastructure and hilly environment. The scalability of these technologies is also assessed. Third, it evaluates the feasibility and performance of the system through an experimental smart bin prototype, demonstrating the potential to reduce operational costs and optimize collection routes.

Following this introduction (Sect. 1), this paper is structured as follows: Sect. 2 presents related work; Section 3 details the methodology used; Section 4 presents the results obtained; Section 5 discusses these results; and Section 6 concludes this paper and outlines future perspectives. It should be noted that this study focuses on architectural design and technical validation; comprehensive statistical analysis of waste generation patterns will be addressed in a subsequent study using data from our deployment.

2 Related works

Several research projects have already been dedicated to smart waste management using the IoT worldwide, more specifically in the context of smart cities. In this section, we present a summary of the most relevant work on:

(1) IoT solutions for smart waste management and (2) applications in developing countries similar to the context of the city of Bukavu in the Democratic Republic of Congo.

2.1 IoT solutions for smart waste management

Smart waste management systems vary globally.

In India, [12] proposed an intelligent waste management system based on a hybrid architecture, combining LoRaWAN for public bins and WiFi for household bins. Although this study demonstrates the technical feasibility of such an approach with a pilot of 16 bins, some critical technical limitations can be identified from their design.

First, the system's reliance on a WiFi network for household bins poses a significant constraint for large scale deployment in a city like Bukavu, where 4 G connectivity is limited and home WiFi penetration is low. Second, the sensor's energy autonomy, even with a solar panel, is reported to be only 70 days without continuous recharging [12], a figure that is insufficient for a sustainable system. Third, although the system integrates a GPS module and a graphical interface for monitoring, it does not include any route optimization algorithm to exploit the collected data for dynamic collection planning.

In Finland, [13] conducted an extensive bibliometric analysis of 173 publications to establish a comprehensive roadmap for smart waste management, identifying key technologies (ultrasonic fill level sensors, GPS, LoRaWAN) adapted to local cost, connectivity, and infrastructure constraints. While this study provides a valuable theoretical framework and maps the key stakeholders (population, municipal services, formal and informal collectors), it remains purely conceptual and does not include any experimental validation or field deployment. Consequently, it offers no quantitative performance metrics, such as battery life, packet delivery ratio, or sensor accuracy, that could be directly compared with real world implementations.

In Malaysia, [14] validated a low cost prototype based on a Raspberry Pi, using a combination of infrared and camera sensors to automatically sort waste into four categories: household, recyclable, hazardous, and non industrial. While the cost effectiveness and the sorting capability are promising, the study reveals significant technical boundaries that limit its applicability for large scale deployment in a city like Bukavu.

First, the reliance on a Raspberry Pi as the processing unit introduces a high power consumption (typically 3 to 5W), making it unsuitable for battery powered, long term deployments in areas with unreliable electricity access. Second, the sensor range is limited to 100–500 cm [14], which is adequate for sorting stations but insufficient for monitoring bins installed in households or public spaces. Third, the study's validation was conducted on a small scale and does not address the network connectivity challenges (WiFi dependency) or the economic sustainability of a city wide deployment.

In Qatar (2024), [15] conducted a sophisticated multi agent simulation study based on real data from the Al Rayyan region (Doha). The system integrates three types of sensors (ultrasonic, weight, density) and uses LoRaWAN, WiFi, and cellular networks (GSM/4 G/5 G) for data transmission. The simulation incorporates advanced optimization algorithms, including Multiple Traveling Salesman Problem (MTSP) and Integer

Linear Programming (ILP), to model collection routes. The results demonstrate a 51% reduction in collection travel distance compared to traditional fixed schedule collection.

While this study represents a significant scientific contribution by simulating IoT impact prior to implementation, some critical limitations must be noted. First, the results are based exclusively on simulation and have not been validated through field deployment. The claimed 51% reduction remains theoretical until confirmed in real world operational conditions. Second, the technical complexity of the multi agent simulation requires substantial local expertise, robust infrastructure (reliable electricity, internet connectivity), and extensive input data (road networks, demographics, waste generation patterns), conditions that are particularly challenging to meet in a city like Bukavu, where electricity access is only 17.8% [16] and network connectivity is intermittent. Third, the study employs multiple communication protocols (LoRaWAN, WiFi, GSM/4 G/5 G), introducing architectural complexity that may not be economically viable or technically sustainable in resource constrained environments.

2.2 Applications in developing countries

Several studies address smart waste management in resource constrained contexts, yet they exhibit significant differences in scope, technical choices, and validation maturity that limit direct comparability.

In South Africa, [17] proposes a comprehensive architecture combining redundant ultrasonic sensors, an ATmega328P microcontroller, and a GSM module for data transmission to a control center. However, this study remains limited by the absence of any field validation, leaving its real world feasibility and robustness unproven.

In Nigeria, [18] adopts a more systemic approach by integrating smart bins into a centralized web platform based on a 4 layer IoT architecture, using HC-SR04 sensors, an Arduino Uno R3, a 16x2 LCD screen, and an ESP8266 WiFi module for real time tracking via unique IP addresses. Unlike the South African study, this work demonstrates economic feasibility (unit cost 23 US dollars) and assesses user acceptability through surveys. However, its reliance on WiFi, explicitly acknowledged as unsuitable for city wide coverage, constitutes a major technical boundary that limits large scale deployment, whereas GSM, despite its higher latency and cost, offers broader coverage.

In Zambia, [19] represents a significant methodological advance by using Agile methodology across four sprints (hardware and calibration, communication, applications, optimizations) to systematically transition from a laboratory prototype to a deployed and operational system. This addresses a critical gap left open by the previous studies, which did not propose any deployment methodology. Nevertheless, the experiment was restricted to a single university campus over six weeks, raising questions about its generalizability to larger, more heterogeneous urban settings.

Finally, the Casablanca study [20] goes further by demonstrating, through complete systemic integration of IoT architecture (Arduino Mega 2560, ultrasonic and PIR sensors, ESP8266, standard 5V power supply) with an open source optimization tool (Open Door Logistics), significant economic and environmental benefits, adding a dimension absent from the other studies. However, it inherits the same WiFi limitation as the Nigerian study.

Critically, across all four studies, no comparative performance metrics (e.g., energy consumption, sensor accuracy, or network reliability) are provided, making it difficult to objectively assess which architecture is most suitable for a given context.

Moreover, none of these studies has been conducted in the eastern part of the Democratic Republic of Congo, particularly in Bukavu, a gap that fully justifies the originality and relevance of the present study, which will address these underexplored dimensions.

2.3 Positioning of this study

This study addresses the gaps identified in the existing literature through six key contributions, while also offering a comparative perspective against recent implementations in sub Saharan Africa and North Africa.

First, it uses EM310-UDL sensors with a battery life exceeding 10 years, directly addressing the limited energy autonomy (70 days) reported in the hybrid system [12] and the high power consumption of Raspberry Pi based solutions [14]. In contrast, comparable studies in South Africa [17] and Nigeria [18] rely on ATmega328P and Arduino based architectures without providing any empirical energy consumption metrics, while the Casablanca project [20] uses standard 5V power supplies, leaving energy autonomy unquantified.

Second, a 12 week field pilot was conducted across three neighborhood types (residential, dense, peri-urban) with 42 households to empirically validate the system under real world constraints, moving beyond the purely conceptual framework proposed for Finland [13] and the simulation based results from Qatar [15]. This pilot also surpasses the six weeks, single campus deployment in Zambia [19] and directly addresses the absence of field validation noted in the South African proposal [17], providing a more robust and heterogeneous validation environment.

Third, the LoRaWAN protocol, chosen for its long range communication capabilities and as a simpler alternative to hybrid architectures [12, 15], was empirically validated in Bukavu's mountainous terrain. This contrasts sharply with the WiFi dependent solutions in Nigeria [18] and Casablanca [20], which are explicitly acknowledged as unsuitable for city wide coverage, and the GSM based approach in South Africa [17], which, while offering broader coverage, incurs higher latency and operational costs.

Fourth, the OpenRemote platform integrates a multi criteria route optimization algorithm, enabling a shift from fixed schedules to a demand driven collection model, a feature absent from the study in Finland [12]. This optimization capability aligns with the Open Door Logistics tool used in Casablanca [20], but our implementation is uniquely grounded in empirical radio link data rather than assumed network conditions.

Fifth, the analysis relies on empirical field data, not simulations [15], with concrete metrics on battery consumption and network reliability. This fills a critical gap common to all four comparator studies (South Africa [17], Nigeria [18], Zambia [19], and Casablanca [20]), none of which provide comparative performance metrics such as energy consumption, sensor accuracy, or network reliability, making objective architectural assessment difficult.

Finally, a comprehensive economic analysis (CAPEX/OPEX) and scalability assessment are provided, grounded in empirical radio link measurements (RSSI/SNR), offering a more pragmatic pathway than the complex, multi protocol architecture proposed in Qatar [15]. While the Nigerian study [18] demonstrates economic feasibility at a unit

Table 1 Comparative analysis of smart waste management IoT solutions

Study	Protocol	Battery life	Route optimization	Validation	Cost data
[12]	Hybrid (LoRa+WiFi)	70 days	No	16 bins	–
[13]	Theoretical (LoRa+GPS)	–	No	Conceptual	–
[14]	WiFi (Raspberry Pi)	3-5W (high)	No	Prototype	Low cost
[15]	Multi (LoRa/WiFi)	–	Yes (ILP/MTSP)	Simulation	–
[17]	GSM	–	No	No field validation	–
[18]	WiFi (ESP8266)	–	No	Surveys	$23
[19]	–	–	No	6 weeks, 1 campus	–
[20]	WiFi (ESP8266)	5V supply	Yes(Open Door)	Partial	–
Our study	LoRaWAN	10 years	Yes (OpenRemote)	12 weeks, 42HH, 3 zones	CAPEX/ OPEX



Fig. 1 EM310-udl and DF703 sensors

cost of 23, our analysis extends this by including operational expenditures and scalability projections based on real world terrain and network performance, thus providing a more complete cost benefit framework for decision makers in similar resource constrained contexts.

To enable a critical and quantitative comparison, Table 1 synthesizes the technical boundaries and reported performance metrics of reference studies against our system.

3 Materials and methods

3.1 General system description

The system developed in this study is based on an experimental approach that involves prototyping and the use of the Internet of Things technology for the physical collection of waste data in the city of Bukavu (pilot case).

The architecture comprises four main components. First, ultrasonic sensors are installed on waste bins to measure fill levels. Second, a LoRaWAN gateway collects data from these sensors. Third, these data are transmitted to a server on The Things Network (TTN) for temporary storage. Finally, the OpenRemote IoT platform is used for data storage, analysis, and visualization.

3.2 Hardware requirements

3.2.1 Sensors

Two sensor models were evaluated: DF703 and EM310-UDL (Fig. 1). The EM310-UDL was selected for its greater detection range (450 cm vs. 400 cm), increased angular

accuracy (60° vs. 2°), extended battery life (10 years vs. 8 years) and lid open or close detection capability via integrated accelerometer. Both use ultrasonic technology, which offers higher field of view and lower cost than infrared, though accuracy can be affected by climate changes and sound-absorbing materials [21].

3.2.2 Gateway

A gateway is an electronic device that acts as an intermediary between the nodes (sensors) and the network server in an IoT infrastructure [22]. On one hand, it receives the modulated LoRa signal [23], and, on the other, transmits the received LoRa frames to the network server via the Internet (using 3 G, 4 G, 5 G, LTE-M, Wi-Fi, or Ethernet connections) [22]. In this study, we opted for a Things Indoor Gateway (see Fig. 4), compatible with EM310-UDL sensors.

3.2.3 Servers

A server is an electronic device responsible for managing and securing data flows on a network [23]. In this study, two types of servers are used for data management and security: the LoRaWAN network server and the application server. The network server handles authentication, de-duplication, and frame transmission, while the application server manages decryption and message delivery via the MQTT protocol to the OpenRemote IoT platform.

3.3 Connectivity technology choice

Three primary factors influence the choice of a communication protocol for an IoT device [23]: the desired range, bandwidth, and energy consumption.

3.3.1 Factors influencing choice

1. **The range** The city of Bukavu covers a total area of 62 km^2, with 43 km^2 of land and 19 km^2 of water [24]. With an average altitude between 1463 and 2200 m [4],the most suitable communication protocol for this city would be one offering extended range to facilitate data exchange between sensors, gateway and server.
2. **Energy** Access to electricity remains a critical challenge in the Democratic Republic of the Congo, particularly in cities such as Bukavu, where only 17.8% of household energy needs are currently met [16]. Therefore, the IoT deployment in the city of Bukavu requires communication protocols with low energy consumption, implemented on devices powered by long-life batteries.
3. **Bandwidth** The city of Bukavu frequently experiences bandwidth limitations and network disruptions, compounded by high communication costs for a population with low-income levels. The most suitable communication protocol for Bukavu would be a long-range protocol with low data rates and an openly accessible frequency band. Based on the above requirements, LPWAN (Low Power Wide Area Network) technologies, which include communication protocols such as LoRaWAN (Long Range Wide Area Network), NB-IoT (Narrowband Internet of Things), etc. [23], appear to be better suited to this city.

Table 2 LoRaWAN and NB-IoT protocols features comparison. Cost estimates are based on typical module prices, connectivity fees, and infrastructure requirements for a deployment of 2,502 sensors over 10 years (see Sect. 5.5)

Features	Protocol	
	LoRaWAN	NB-IoT
Network range	2 to 5 km (urban area)	1 km (urban area)
	10 to 20 km (rural area)	10 km (rural area)
Battery life	> 10 years	10 years
Bandwidth	300 bytes to 50 kilobytes per second	200 kilobytes per second
Cost[a]	Weak	High

[a] LoRaWAN has low module and operational costs; NB-IoT requires higher module costs and yearly cellular fees. Over 10 years with 2,502 sensors, LoRaWAN is 5–10 times less expensive

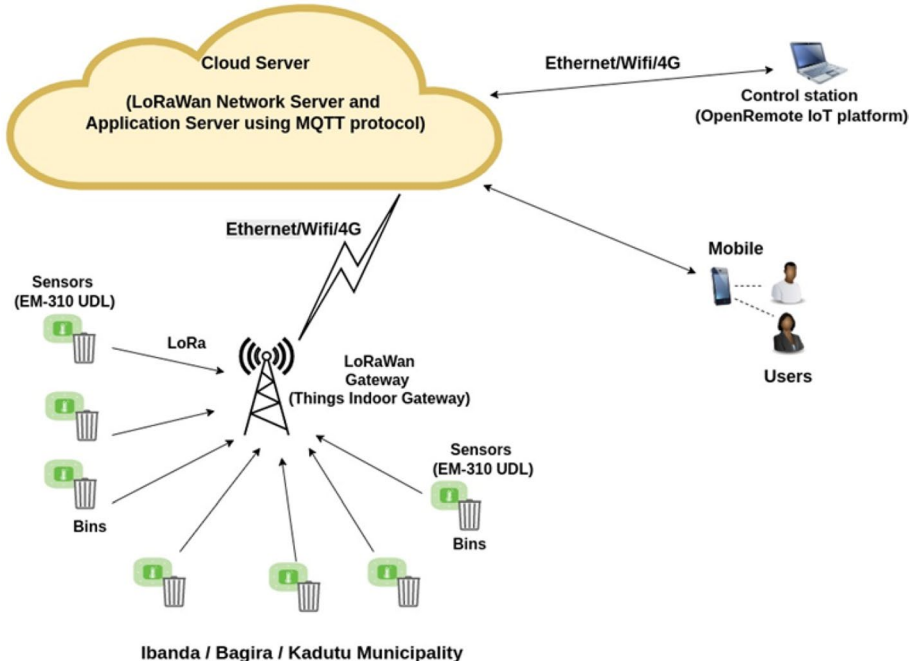


Fig. 2 Proposed network architecture for the city of Bukavu

3.3.2 LoRaWAN versus NB-IoT protocols: selection for Bukavu

LoRaWAN (Long Range Wide Area Network) is a low speed, long range, very low power protocol [23] with free frequencies (433 MHz in Africa/Asia [25]). NB-IoT is its main competitor within LPWAN [26]. Table 2 compares their features.

LoRaWAN was selected for Bukavu because it offers wider network coverage, lower energy consumption, lower operational cost (no cellular fees), and uses freely accessible frequency bands. While NB-IoT provides higher bandwidth, waste monitoring requires low data rates, making LoRaWAN the appropriate choice for Bukavu's constraints.

3.4 Proposed network architecture

Therefore, the following network architecture is proposed in this study (Fig. 2):

Equipped with a LoRa radio transmission module, the endpoints (EM310-UDL ultrasonic distance sensors) collect information on the status of the waste bins and transmit it to a LoRaWAN gateway (Things Indoor Gateway).

This gateway, featuring two communication interfaces (LoRa radio and Internet), forwards the received frames to a LoRaWAN network server configured on The Things Network (TTN). This server handles authentication and deduplication.

The processed data is then passed to the application server. After decrypting the messages, this server sends them via the MQTT protocol over an Internet connection (4 G or Wi-Fi) to the OpenRemote IoT platform, named "Bukavu Waste Management System," for final processing. Users subsequently access the data via the platform's dashboard for visualization and final decision-making.

The above LoRaWAN network architecture does not impose a strict limit on the sensors numbers that a gateway can support, as it does not manage addressing directly but rather passively listens to all radio packets passing through it.

The security implications of this architecture, including key management and vulnerability to attacks, are analyzed in Sect. 5.6.

3.5 Extended form of the proposed architecture

The extended form of the proposed network architecture in this study, was derived using a simple geometric modeling approach, commonly employed in radio engineering [27]. This approach assimilates each gateway's coverage area to a disk with an effective radius [27], simplifying node placement and quantity calculations. The minimum number of nodes N, is therefore given by:

$$N = \left\lceil \frac{A_{city}}{\pi R_{eff}^2} \right\rceil \tag{1}$$

Where,

- A_{city}: city area (in km^2)
- R_{eff}: city effective radius coverage (in km)

Considering the radius coverage and fixing sensibility, empirical measurements conducted in a dense urban context and similar to Bukavu are invaluable. A study conducted in the city of Oulu, Finland [28], reports that with high parameters (SF12 and 14 dBm), the LoRaWAN network maintains a transmission reliability greater than 85% up to a distance of 5 km. Given the geographical context of Bukavu [4], adopting a coverage radius significantly lower than the maximum range reported in the study [28] would provide a safety margin for experimental uncertainties and potential variations in operating conditions.

The Bukavu geographical data context, the LoRaWAN networks range in urban environments (2 to 5 km) [26], as well as local measurements reported under similar conditions [29], all converge towards adopting a coverage radius that guarantees satisfactory service quality in a complex environment. Hence, the choice of an effective coverage radius of 1.5 km for the modeling in this study, slightly below the minimum range of LoRaWAN networks discussed in study [26].

Applied to our case, the minimum number of the required nodes is given in Equation 2:

$$N = \left\lceil \frac{62}{3.14 \times 1.5^2} \right\rceil = \lceil 8.779 \rceil = 9 \tag{2}$$

Therefore, to ensure comprehensive and robust radio coverage for the city of Bukavu, considering environment, a minimum deployment of 9 LoRaWAN gateways is required. This preliminary estimate should be subsequently refined through a modeling phase involving a propagation model adapted to the mountainous terrain, coupled with an optimization algorithm, to determine the optimal spatial configuration of the gateways.

Based on the constraints identified above, the extended network architecture for the city of Bukavu (Fig. 3), should thus consist of 9 Internet-connected LoRaWAN gateways and several thousand EM310-UDL sensors [30]. To address the power supply deficit [16], these gateways should be equipped with batteries coupled with solar panels. Their radio sensitivity must be high [31], to ensure coverage adapted to the city's mountainous terrain.

A LoRaWAN server hosted on The Things Network (TTN), an application server that receives TTN data, and an open-source IoT platform, such as OpenRemote, are also required for data management. Finally, 4 workstations for activity monitoring, with access to the IoT platform's dashboard, must be integrated into this architecture, corresponding to one workstation per municipality.

3.5.1 Scalability analysis and network capacity assessment

To validate the applicability of the extended network architecture, a comprehensive scalability analysis was conducted. This assessment evaluated the architecture's capacity to support a large-scale deployment in the city of Bukavu.

To evaluate the scalability of our system, we adopted the network capacity model defined in [32]. In this model, the maximum capacity of a LoRaWAN gateway ($O_{\max}$) represents the highest offered traffic (in packets/day) it can handle while maintaining a success rate (P_s) above a required minimum threshold ($P_{s,\min}$).

The maximum number of sensors supported by the network can be expressed as:

$$N_{\max} = \frac{O_{\max}}{\lambda} \tag{3}$$

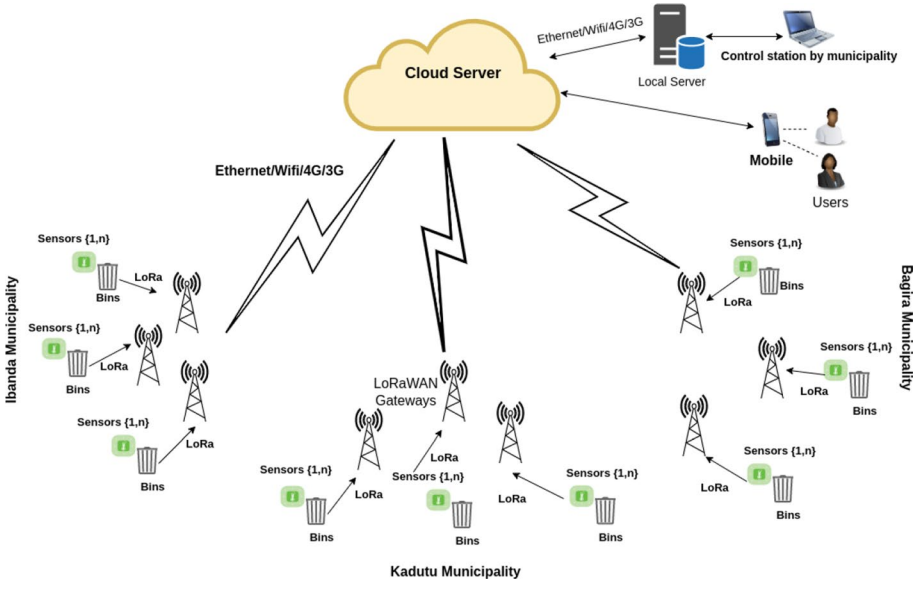


Fig. 3 Extended network architecture of Bukavu city

where:

- N_{max} = Maximum number of deployable sensors
- O_{max} = Network capacity (determined by simulation or modeling)

1. λ = Packet generation rate per sensor (packets/day)**System Configuration Parameters**

   Consistent with the configuration described in Sect. 3.6.1, our EM310-UDL sensors transmit measurements at regular 10-minute intervals. The packet generation rate per sensor (λ) is therefore derived as:

$$\lambda = \frac{24 \times 60}{10} = 144 \text{ packets per day per sensor} \tag{4}$$

   The messages transmitted by the sensors include fill level (measured distance), battery status, and cover state (open/closed). Consequently, the typical payload is less than 50 bytes, categorizing these as small data packets comparable to the use cases in Group A of the taxonomy defined in [32] (Waste Management, PL = 10-20B).
   For waste management applications, sporadic packet loss is tolerable because bin status is updated every 10 min. We therefore set the minimum required success rate (Ps,min) to 90%, consistent with the requirements for similar applications [32].

2. **Theoretical capacity of a single gateway**

   Considering the deployment of a single gateway covering a district of Bukavu (e.g., Ndendere, Nyakavogo), and based on the simulation results from [32] for low-payload applications (Group A) with an optimal SF7 configuration, the network capacity for a single LoRaWAN gateway (O_{max}) is estimated at approximately 390,000 packets per day.
   Hence, the maximum number of sensors supported by a single gateway under these ideal conditions is:

$$N_{max} = \frac{390000}{144} = 2708 \text{ sensors} \tag{5}$$

   Thus, a single LoRaWAN gateway strategically placed within a district can theoretically support up to 2708 sensors within its optimal coverage radius, provided that all sensors can operate using a low spreading factor (SF7).
   However, as observed in our field measurements (Table 6), several sensors exhibited degraded radio link quality indicators (RSSI ≤ -107 dBm, negative SNR) and consequently utilized higher SFs (SF9, SF10) to maintain connectivity.
   According to [32], the use of high SFs (greater than or equal to 9) significantly reduces network capacity due to increased time on air. This capacity can drop to as low as 40,000 packets/day (large zone scenario, Group C).
   This reduction would bring the number of sensors supported by a single gateway down to:

$$N_{\max} = \frac{40000}{144} = 278 \text{ sensors} \tag{6}$$

Consequently, this sharp decline highlights the scalability challenges anticipated in areas of Bukavu where radio coverage is limited.

3. **Capacity with multiple gateways**

The analysis conducted in Sect. 3.5 established that the minimum number of gateways required to ensure complete radio coverage for the city of Bukavu is 9. The maximum number of sensors supported by the entire gateway infrastructure can be expressed as [32]:

$$O_{\max,\text{total}} = 9 \times \alpha \times O_{\max,1} \tag{7}$$

where α represents a gain factor associated with spatial diversity and collision reduction, estimated to range between 3 and 5 based on the findings in [32].
Thus, considering $O_{\max,1} = 390000$ packets/day for a single LoRaWAN gateway in an optimal configuration, and adopting a conservative gain factor (α) of 3 for 9 gateways, we obtain:

$$O_{\max,\text{total}} = 9 \times 3 \times 390000 = 10530000 \text{ packets/day} \tag{8}$$

The total number of sensors ($N_{\max,\text{total}}$) supportable citywide is then given by:

$$N_{\max,\text{total}} = \frac{O_{\max,\text{total}}}{\lambda} \tag{9}$$

Thus:

$$N_{\max,\text{total}} = \frac{10530000}{144} = 73125 \text{ sensors} \tag{10}$$

In summary, in Bukavu areas with limited radio coverage, the sensor density per gateway must be limited to approximately 278 sensors. This constraint arises because such areas require higher spreading factors to maintain connectivity, in contrast to well-covered zones where the required sensor density per gateway can reach up to 2708 sensors using low spreading factors (SF7).

3.6 Prototype configuration and deployment

3.6.1 Prototype configuration

In this study, all devices were configured with a frequency plan between 863 and 870 MHz and a spreading factor (SF) that varied between (SF) 9 and 12, to achieve a good balance between range and data throughput in LoRa communication.

3.6.1.1 Cloud server configuration We configured the cloud server following the standard TTN registration and API integration procedure[28, 29].



Fig. 4 Things indoor gateway

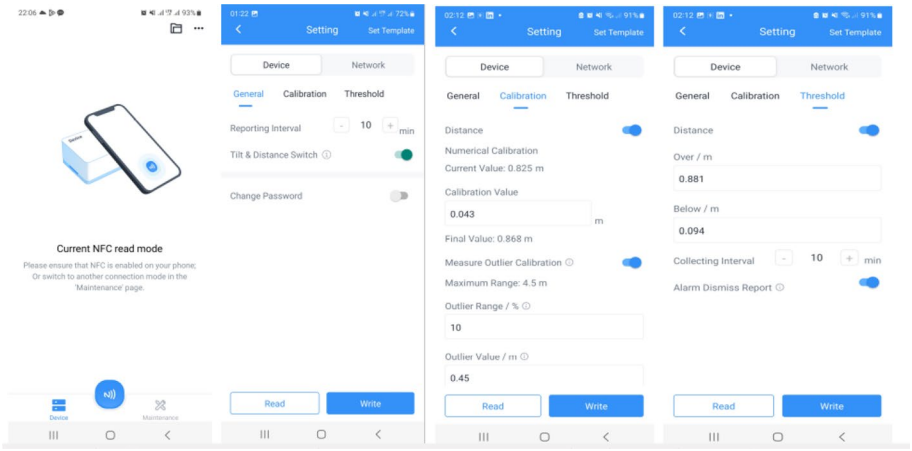


Fig. 5 Sensor calibration

3.6.1.2 LoRaWan Gateway Configuration Due to its low cost and technical suitability for deployment in Bukavu, we selected the "Things Indoor Gateway" (Fig. 4). Although designed for indoor use, its range of 2 to 3 km in urban areas [26] makes it adequate for neighborhood level coverage.

The configuration and registration process on The Things Stack network followed these steps: identifier generation, power and activation, network connection and platform registration.

3.6.1.3 Sensors configuration Sensors were configured using a hybrid calibration strategy (two-point, digital, and outlier rejection) and mounted on bin lids facing downward (Fig. 5). A 10 min reporting interval was selected, with empty and full states defined at 881 mm and 94 mm, respectively [6, 31, 33–35].

3.6.1.4 IoT platform configuration OpenRemote was selected as the IoT platform due to its open source model, broad protocol support (MQTT, HTTP, Zigbee), automation capabilities, and active community [36]. The platform was deployed using Docker containers, and map tiles were customized for Bukavu following the official documentation [29].

3.6.1.5 TTN and OpenRemote connection via MQTT A secure MQTT link was established between the deployed sensors and the local OpenRemote instance, allowing two-way communication.

Docker was reconfigured to use the workstation's network interface for MQTT communications.

A service user with read/write privileges was configured within the project domain, and MQTT Explorer was used as the client to establish secure, bidirectional communication between the sensors and the OpenRemote platform (Fig. 6).

3.6.1.6 Agent and assets configuration in OpenRemote The OpenRemote platform was integrated with the LoRaWAN network via The Things Stack using an MQTT agent, and physical EM310-UDL sensors were modeled as digital assets with battery, distance, and position attributes [30, 36, 37].

3.6.1.7 Smart rules configuration To provide the IoT system with advanced decision making capabilities, several smart rules were created within the OpenRemote platform, using the Groovy language [38]. These rules implement four main algorithms that interact to optimize waste management.

3.6.2 Algorithm 1: Bin fill level monitoring and forecast

The bin fill level monitoring algorithm (Supplementary material S2.1) integrates real-time data acquisition, analytical processing, and alert management. When the lid is open, the distance measurement is discarded because the ultrasonic beam no longer points vertically into the bin. The system retains the last valid fill percentage until a new valid reading is obtained. This ensures fill level estimates are based only on measurements taken when the lid is properly closed. The parameters used in this algorithm are provided in Supplementary material ST2.

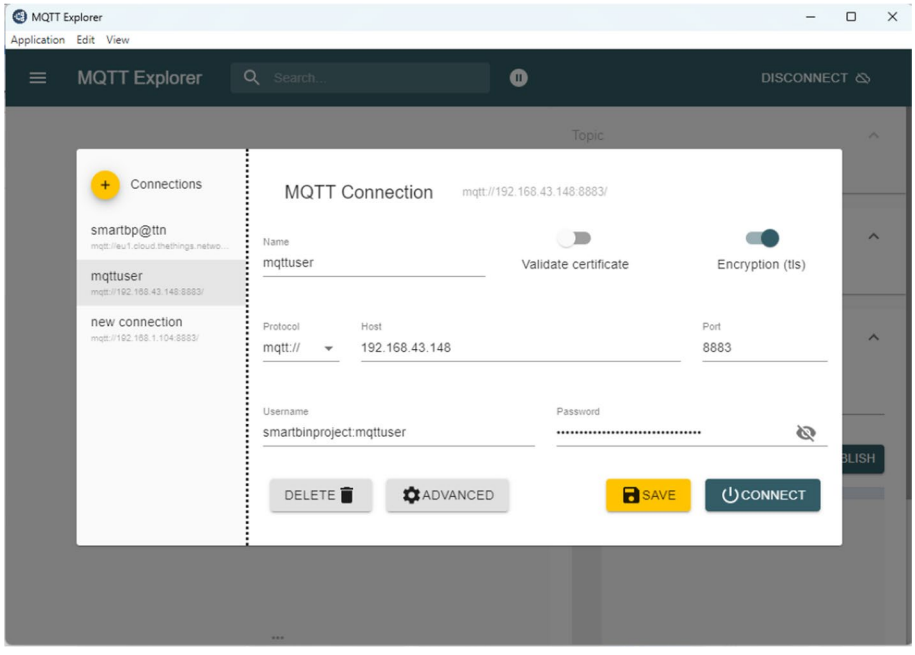


Fig. 6 Connecting the MQTT Explorer broker to the MQTT server integrated in TTN

The following mathematical formulas are used in this algorithm:

$$f_i = \frac{emptyDist - d_i}{emptyDist - fullDist} \times 100, \tag{11}$$

where *emptyDist* and *fullDist* denote the distance values measured for empty and full bins, respectively, d_i is the current distance measurement, and f_i represents the bin filling percentage.

$$V_i = \frac{f_i}{100} \times volume, \tag{12}$$

where V_i is the estimated waste volume in bin *i*.

$$W_{waste} = V_i \times density, \tag{13}$$

where W_{waste} represents the waste weight and *density* represents the configured waste density.

$$W_{\text{total}} = W_{\text{waste}} + W_{\text{empty}}, \tag{14}$$

where W_{total} denotes the total bin weight and W_{empty} is the configured empty bin weight.

$$t_{full} = t_i + \left(\frac{90 - f_i}{r_i}\right) \times 3600, \tag{15}$$

where t_{full} represents the estimated time at which the bin reaches the critical filling threshold (90%), t_i is the current time, f_i is the current filling percentage, and r_i denotes the hourly filling rate.

3.6.3 Algorithm 2: Behavioral data collection

The behavioral data collection algorithm (Supplementary material S2.2) complements the bin filling prediction system by providing a behavioral analysis of usage patterns to optimize waste collection. Supplementary material ST3 provides the full set of this algorithm's parameters.

The deposit efficiency and behavioral score calculations are based on the following mathematical formulas:

$$depositEfficiency = \min\left(100,\ \frac{deposits}{dOpenings} \times 100\right), \tag{16}$$

where *depositEfficiency* denotes the deposit efficiency, *deposits* the number of detected waste deposits, and *dOpenings* the number of daily cover openings.

$$bScore = 100 - \max(0,\ dO - 20) \times 2 - nO \times 5 - \max(0,\ 50 - dE) \times 0.5 \pm A_{fill} \tag{17}$$

where *bScore* represents the behavioral score, *dO* and *nO* represent daily and nightly openings, *dE* is the deposit efficiency, and A_{fill} is the fill factor.

3.6.4 Algorithm 3: Intelligent waste collection planning

The intelligent waste collection planning algorithm (Supplementary material S2.3) complements the previous algorithms by implementing an intelligent waste collection

planning system, driven by both event-based and periodic monitoring. Its parameters are available as supplementary material ST4.

It integrates the actual fill level of waste bins with historical collection data to optimize scheduling. The following mathematical formulas are used in this algorithm:

$$fill_{pct} = \max\left(0,\ \min\left(100,\ \text{round}\left(\frac{emptyDist - d}{emptyDist - fullDist} \times 100\right)\right)\right) \tag{18}$$

where $fill_{pct}$ denotes the bounded and rounded bin filling percentage.

$$Urgency = \begin{cases} \text{"Critical"}, & \text{if } fill_{pct} \geq T_{critical} \\ \text{"High"}, & \text{if } T_{high} \leq fill_{pct} < T_{critical} \\ \text{"Low"}, & \text{if } fill_{pct} < T_{high}, \end{cases} \tag{19}$$

where the urgency level is classified as *Critical*, *High*, or *Low* depending on the bin filling percentage relative to the defined thresholds.

$$next_{collection} = \begin{cases} last_{collection} + Critical_{delay}, & \text{if Urgency = "Critical"} \\ last_{collection} + High_{delay}, & \text{if Urgency = "High"} \\ last_{collection} + Low_{delay}, & \text{if Urgency = "Low"}, \end{cases} \tag{20}$$

where $next_{collection}$ is the estimated next collection date, $last_{collection}$ corresponds to the last collection date, $Critical_{delay}$ refers to the critical collection interval delay (3 days), $High_{delay}$ evokes the high collection interval delay (7 days), and Low_{delay} corresponds to the lowest collection interval delay (14 days).

3.6.5 Algorithm 4: Waste collection routes optimization

The waste collection routes optimization algorithm (Supplementary material S2.4) implements a system for optimizing waste collection routes, incorporating constraints derived from the road network of the city of Bukavu. Its parameters are provided in supplementary material ST5.

4.1. Fundamental calculations

$$P(b_i) = \beta(u_i) + \alpha \cdot f_i \tag{21}$$

where:

- $P(b_i)$: overall priority associated with bin b_i
- $\beta(u_i)$: emergency weighting function (100 if "Critical", 80 if "High", 50 if "Medium", 0 if "Low")
- f_i: bin fill percentage (0-100)
- α: weighting coefficient

$$d_{\text{haversine}}(p, q) = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos\phi_1 \cdot \cos\phi_2 \cdot \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right) \tag{22}$$

where R is the Earth's mean radius (6371 km).

$$d_{\text{road}}(p, q) = d_{\text{haversine}}(p, q) \times \delta(p, q) \tag{23}$$

where $\delta(p, q)$ is the detour factor.

$$t(p,q) = \frac{d_{\text{road}}(p,q)}{v_{\max}(r(p,q))} \times T(r(p,q)) \times 60 \tag{24}$$

where $v_{\max}$ is the theoretical maximum speed and T is the speed reduction factor.

$$T_{\text{route}} = t(D,b_1) + \sum_{i=1}^{k-1} t(b_i,b_{i+1}) + t(b_k,D) \tag{25}$$

where D is the central depot.

$$S(p,b_j) = \alpha_d\, d_{\text{road}}(p,c_{b_j}) + \alpha_t\, t(p,c_{b_j}) - \alpha_p\, P(b_j) \tag{26}$$

where $S(p,b_j)$ evaluates the attractiveness of bin b_j from position p.

4.2. Mathematical formulation

The route optimization problem is formulated using the equations and parameters already defined in this section:

- Bin priority $P(b_i)$ is given by Equation 21: $P(b_i) = \beta(u_i) + \alpha \cdot f_i$.
- Road distance $d_{\text{road}}(p,q)$ is given by Equation 23: $d_{\text{road}}(p,q) = d_{\text{haversine}}(p,q) \times \delta(p,q)$.
- Travel time $t(p,q)$ is given by Equation 24: $t(p,q) = \frac{d_{\text{road}}(p,q)}{v_{\max}(r(p,q))} \times T(r(p,q)) \times 60$.
- Total route time T_{route} is given by Equation 25: $T_{\text{route}} = t(D,b_1) + \sum_{i=1}^{k-1} t(b_i,b_{i+1}) + t(b_k,D)$.
- Attractiveness score $S(p,b_j)$ is given by Equation 26: $S(p,b_j) = \alpha_d\, d_{\text{road}}(p,c_{b_j}) + \alpha_t\, t(p,c_{b_j}) - \alpha_p\, P(b_j)$.
- Parameters $\alpha_d = 0.3$, $\alpha_t = 0.4$, $\alpha_p = 0.3$, and truck capacity $Q = 2000$ kg are defined in Supplementary material ST5.

The optimization aims to select a set of routes starting and ending at the depot $D(-2.4984, 28.8482)$ such that:

- Each bin requiring collection (urgency "Critical" or fill $\geq 90\%$) is visited exactly once,
- The total estimated waste weight per route does not exceed Q,
- The total travel time summed over all routes (Equation 25) is minimized.

The ordering of bins within a route is guided by the attractiveness score (Equation 26), which combines distance, time, and priority.

4.3. Computational complexity analysis

The current manuscript does not include a formal computational complexity analysis of the route optimization algorithm, as the primary focus of this study is the architectural design and technical validation of the IoT system. The pilot deployment involved at most $n = 7$ bins requiring collection at any time (Sect. 3.7). On the Intel Core i5-3360 M 4 GB RAM workstation used for the pilot (see Sect. 3.6.1), the optimization completed within acceptable time; however, precise runtime measurements were not recorded. The reported "estimated time" (e.g., 13 min in Sect. 4.3.4) refers to the projected travel duration for the collection vehicle, not the computational time of the algorithm. A detailed

complexity analysis, including worst-case bounds with respect to the road network size and the number of bins, is beyond the scope of this paper and is left for future work.

From the equations above, computing each attractiveness score (Equation 26) requires a road distance calculation (Equation 23) and a travel time calculation (Equation 24). For a set of n bins, a naive implementation would cost $O(n^2)$ such computations. In the pilot ($n \leq 7$) this is trivial; for larger deployments, the distance matrix can be precomputed because bin positions are fixed.

3.6.5.1 Application dashboards configuration Data visualization for the developed IoT system is provided by a series of dashboards designed within OpenRemote Insights module [37]. Some results corresponding to these dashboards are presented in Sect. 4.

3.6.6 Prototype deployment

To empirically validate the proposed IoT architecture, a pilot deployment of seven smart bin prototypes was conducted in Bukavu city. This section outlines the deployment methodology designed to ensure result reproducibility and to validate the proposed optimization strategy against real world operational constraints, including road infrastructure, traffic regulations (speed limits), and the load capacity of collection vehicles.

1. **Depot location and vehicle capacity**

   The central waste depot for our pilot project is located in the municipality of Kadutu with the following geographic coordinates: −2.4984, 28.8482. The waste collection vehicle used in the project has a maximum capacity of 2000 kg (2 tons) of waste (truck capacity constraint).

2. **Bukavu road network representation**

   Bukavu's road network was categorized into three types based on the OpenStreetMap classification [39]: primary (main arteries), secondary (connecting roads), and residential (local streets). Each road type (Supplementary material S4) was assigned specific traffic and detour factors for the route optimization algorithm.

3. **Road network parameters**

   To reflect Bukavu's urban morphology, traffic and detour factors from the route optimization algorithm were applied (Supplementary material ST5). Traffic factors account for delays from intersections, road conditions, and congestion across road types. Detour factors represent the ratio of actual road distance to straight-line Haversine distance. We adopted a traffic factor of 1.2 for primary roads (fewer intersections, smoother flow), 1.5 for secondary roads (narrower, more intersections, denser traffic), and 1.8 for residential roads (narrow, sometimes poor condition, many turns, pedestrians, children). Detour factors were set at 1.6 for Ibanda municipality (dense historical and commercial center [40]), 1.4 for Kadutu (transition zone), and 1.2 for Bagira (more airy peri-urban area).

4. **Detour factor calculation**

The detour factor $\delta(p, q)$ accounts for the difference between the straight-line Haversine distance [41] and the actual road distance due to Bukavu's topography and the circuitous road layout. It is calculated based on the location of the points:

- **Urban center** (within 2 km of coordinates −2.50, 28.87): $\delta = 1.6$
- **Suburban areas** (direct distance less than 3 km): $\delta = 1.4$
- **Periurban areas** (direct distance greater than or equal to 3 km): $\delta = 1.2$

For routes passing through multiple zones, the factor is applied segment-wise. The road distance is then derived from Equation 23:

$$d_{\text{road}}(p, q) = d_{\text{haversine}}(p, q) \times \delta(p, q) \tag{27}$$

For the example route shown in Fig. 16 (Bin 4), the detour factor was 1.6, calculated as 4.5 km/2.8 km, which corresponds exactly to the urban center factor.

5. **Route feasibility validation**

Before returning a route, the algorithm performs the following validation checks:

- Coordinate validation: Each bin's coordinates are checked against the geographic bounds of Bukavu (latitude −2.56308 to −2.45795, longitude 28.76238 to 28.92426). Invalid coordinates trigger a warning and the bin is excluded from optimization.
- Capacity check: The total weight of the bins in each route must not exceed the truck capacity (2000 kg). If exceeded, the route is divided into multiple trips.
- Connectivity check: The algorithm verifies that all bins are reachable via the road network. If a bin appears isolated, it is flagged for manual inspection.
- Reasonable distance check: The calculated route distance is compared to the Haversine distance multiplied by the maximum plausible detour factor (set to 2.0). If the ratio exceeds this threshold, the route is flagged for review.

3.7 Pilot design and deployment timeline

The pilot deployment was designed to evaluate the system's performance in the urban context of the city of Bukavu. We adopted a stratified sampling approach based on three neighborhood types: residential (low density, higher income), dense (high population density, low-income), and peri-urban (transitional zones) [42–44].

3.7.1 Spatial distribution and site selection process

For each deployment phase, the seven smart bins were distributed at increasing distances from the gateway to evaluate network performance under varying link conditions. The selection of specific household locations followed a documented three step protocol: (1) **coverage test** (RSSI and SNR measurements were recorded for two consecutive hours at the candidate location using the procedure described in Sect. 4.2); (2) **accessibility check** (the road segment was verified to be passable by a collection vehicle, width > 2.5 m); (3) **homeowner consent** (written informed consent was obtained). The resulting radio link quality metrics for the Ndendere deployment (Table 6) illustrate the spatial diversity achieved: three sensors (2, 3, 5) operated with RSSI > −100 dBm (closer

to the gateway), while four sensors (1, 4, 6, 7) exhibited RSSI ≤ -107 dBm (further away or with obstructed paths). This radial distribution, replicated in each of the six pilot sites, ensures that the pilot captures the link degradation expected in a city-wide deployment.

3.7.2 Household selection

Households were recruited via local community leaders across pilot neighborhoods. Information sessions informed participants about the study's objectives, deployment duration (up to two weeks), and data collected (fill levels, lid openings, weight estimates).

Participation was voluntary and uncompensated. Selection aimed to capture socio-demographic diversity (family size, economic activity, presence of children) through non probabilistic sampling. Overall, 42 unique households took part, 14 per neighborhood type per phase, all headed by adults aged 18 and over.

3.7.3 Deployment schedule

The 12 week pilot ran from November 10, 2025, to March 28, 2026, across three 4 week phases: Phase 1 in residential neighborhoods (Nyalukemba, Ndendere), Phase 2 in peri-urban areas (Nyakavogo, Lumumba), and Phase 3 in dense neighborhoods (Nyakaliba, Kasali). Each of the 7 bins was deployed in two households per phase for 14 days each, yielding 14 households per phase and 42 unique households overall. This sequential rotation enabled data collection across all neighborhood types with a fixed number of bins. Each phase was preceded by an 8 day sensitization campaign to secure participation, followed by a 3 day network coverage test to verify signal strength. Thus, actual data collection began, on average, 11 days after the start of each preparatory phase.

3.7.4 Bin mobility

All 7 smart bins were deployed simultaneously during each phase and moved to new households at the beginning of each new phase (Fig. 7).

Thus, each bin generated data from all three neighborhood types. The gateway (Things Indoor Gateway) was also relocated to a central location within each neighborhood to



Fig. 7 Bins prior to deployment

Table 3 Deployment locations

Location	Neighborhood	Gateway	Start	End	Days	Prep
Nyalukemba	Residential	Ibanda	10/11/2025	24/11/2025	14	A10, N4
Ndendere	Residential	Ibanda	09/12/2025	23/12/2025	14	A10, N4
Nyakavogo	Peri-urban	Bagira	30/12/2025	13/01/2026	14	A5, N2
Lumumba	Peri-urban	Bagira	24/01/2026	07/02/2026	14	A8, N2
Nyakaliba	Dense	Kadutu	25/02/2026	11/03/2026	14	A12, N5
Kasali	Dense	Kadutu	14/03/2026	28/03/2026	14	A2, N1

A = Awareness days, N = Network coverage test days, Prep = Preparatory period before data collection. Dates in DD/MM/YYYY format. Sensors 1 to 7 were used in all locations

Table 4 Bin identifiers (sensors)

Sensor	Bin ID (TTN / OpenRemote)
Sensor 1	sensor1-24e124713c245522 / 5pxWFBMvwPkl5VV00D0iWX
Sensor 2	sensor2-24e124713c245638 / 2WoApYaLva6nIxiF6nUzGd
Sensor 3	sensor3-24e124713c245733 / 5fBQX5KN7XdRpFWdl6ghNd
Sensor 4	sensor4-24e124713c245243 / 6F2d4jEEKm7NQEM1HlWr3S
Sensor 5	sensor5-24e124713c245531 / 7Yk4VcYiVQqezB1f45A6Fz
Sensor 6	sensor6-24e124713c246554 / 595420IjCBRMQ2IueSzHy5
Sensor 7	sensor7-24e124713c245085 / 3NgwtEUnWFX3wwJgOBl8pn

ensure optimal coverage. Table 3 provides the complete deployment log for all bins, including anonymized location, dates, number of valid measurement days, household awareness and network test coverage days.

Table 4 associates each bin with its identifiers.

3.7.5 Household-level data and security

For each participating household, the system collects the following data from the smart bin:

- **Fill level**: distance measurement converted to percentage fill (0–100%), recorded every 10 min.
- **Lid status**: "normal" (closed) or "tilt" (open), recorded whenever a change occurs (see Sect. 4.1.3).
- **Estimated waste weight**: calculated from fill level and assumed density (0.20 kg/L), as detailed in Sect. 3.6.1.
- **Timestamps**: date and time of each measurement and lid event.
- **Location**: generalized to the neighborhood level (e.g., "Ndendere, Labote Avenue") for analysis; precise GPS coordinates are stored only temporarily for route optimization and are anonymized in all published data.

The system implements multiple layers of security to protect data, complementing the analysis presented in Sect. 5.6:

- **Encryption**: All data transmitted between sensors, gateway, TTN, and OpenRemote are encrypted using TLS, as detailed in Sect. 5.6.4.
- **Authentication**: API keys (98 characters) and service user credentials are required for all data access (Sect. 3.6.1).

- **Access restriction**: Only the research team has access to the OpenRemote platform and the raw sensor data. The platform uses role-based access control via Keycloak, as described in Sect. 5.6.4.

In summary, data collection took place from November 2025 to March 2026 across six sites in Bukavu, representing three neighborhood types (residential, peri-urban, dense) with two sites each. Seven sensors per site (42 total) produced 588 valid sensor days. Preparation required 65 days total, 47 for community outreach and 18 for technical testing, depending on local conditions, and two sites are still active. The database stores no personal identifiers; bin IDs are anonymized, GPS coordinates serve only internal logistics, and all published analyses rely on aggregated household data to ensure privacy.

3.8 Reproducibility protocol

To enable independent replication of this pilot study in other contexts, a full reproducibility protocol (Supplementary material S3) is provided, covering hardware (EM310-UDL sensors), TTN infrastructure, OpenRemote platform, configuration parameters, deployment steps, and adaptation guidelines for other urban settings. To specifically support the reproducibility of site selection, Supplementary Table ST1 offers an anonymised illustrative log for the Ndendere pilot site (Phase 2), based exclusively on radio metrics recorded on December 9, 2025.

4 Results

This section presents results from the 12 week pilot deployment. These results provide evidence for the technical feasibility and operational performance of the proposed IoT architecture.

The collected data are analyzed here primarily as evidence of system functionality rather than as a comprehensive statistical characterization of waste production in Bukavu. Inferential statistical analysis falls outside the scope of this architectural validation paper. Specifically, we evaluate:

- **Sensor performance**: battery autonomy, measurement reliability, calibration accuracy (Sects. 4.1–4.1.2)
- **Network reliability**: RSSI/SNR distributions, packet delivery ratios, connectivity across terrain types (Sect. 4.2)
- **System functionality**: real-time monitoring capabilities, alert generation, behavioral tracking, route optimization demonstration (Sects. 4.1.3 and 4.3)

While the collected data provide preliminary insights into waste generation patterns (e.g., fill rates by neighborhood), we emphasize that inferential statistical analysis, including confidence intervals, ANOVA tests, and predictive modeling, falls outside the scope of this architectural validation paper. These analyses will be the focus of a forthcoming companion study using data from the current deployment.

4.1 Measurements taken by sensors

The EM310-UDL ultrasonic sensors used in our study provide continuous time series data with a measurement frequency of 10 min. They primarily collect data on fill level ("distance_mm"), battery level ("battery"), and bin lid status ("position"). All fill

percentages and derived waste masses presented in this section are estimates based on ultrasonic distance measurements and an assumed waste density. They have not been validated against manual reference measurements and their absolute accuracy is not quantified.

4.1.1 Bin fill level

To better track the steady increase in waste volume, the raw distance measurements collected by the sensors were automatically converted into fill percentages within OpenRemote.

When the fill percentage reaches a threshold of 90% or higher, a critical alert is automatically generated, and an urgent red visual marker is displayed on the dashboard.

The fill percentage is rounded to the nearest integer before comparison with the critical threshold (90%). A bin with an actual fill of 89.5% will be reported as 90% and trigger an alert, while a bin at 89.4% will be reported as 89% and not trigger. This rounding introduces a ±0.5% uncertainty at the threshold boundary.

When combined with the ultrasonic measurement error of approximately ±2.8% at mid-range (derived from the sensor's datasheet accuracy of $\pm(1 + 0.3\% \times S)$ cm, as detailed in Sect. 5.3), the effective threshold range widens to approximately 86.7% to 93.3%. This combined uncertainty remains within acceptable operational limits for waste collection scheduling.

As an illustration, the Fig. 8 shows the variation in the fill percentage of Bin 3 during the first day of collection (Nyalukemba neighborhood, Muhumba Avenue).

The fill percentages for all bins can also be viewed in real time, allowing the progression of waste volume to be tracked (Fig. 9).

4.1.2 Battery evolution per sensor

Prior to the actual deployment of the bins in the Nyalukemba neighborhood in November 2025, specifically on the 10th, a 10 month sensor configuration and testing period



Fig. 8 Change over time in the fill percentage of Bin 3 during the first day of Phase 1 (residential neighborhood, Nyalukemba, Muhumba Avenue, November 10, 2025)

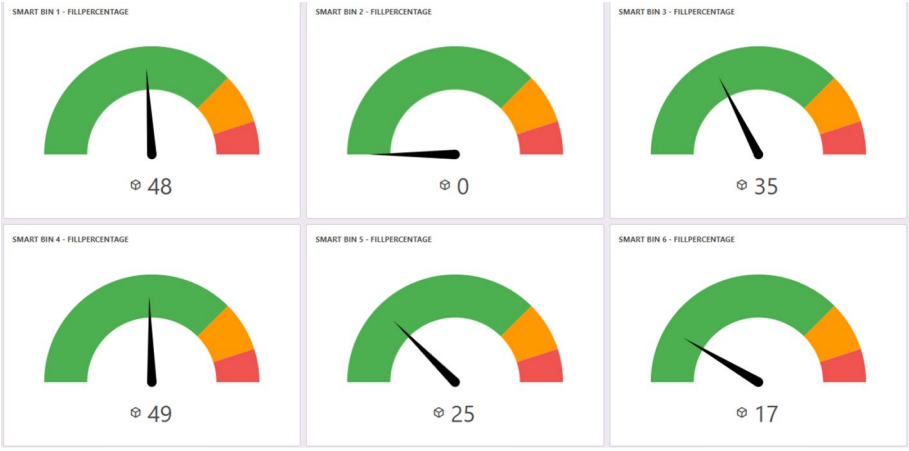


Fig. 9 Percentage of all bins filled in real time

Table 5 Battery evolution per sensor over 11 months

Sensor ID	Initial battery (%)	Actual value (%)	Duration	Obs
sensor1-24e124713c245522	100	83	11	Higher
sensor2-24e124713c245638	100	92	11	Normal
sensor3-24e124713c245733	100	91	11	Normal
sensor4-24e124713c245243	100	92	11	Normal
sensor5-24e124713c245531	100	92	11	Normal
sensor6-24e124713c246554	100	91	11	Normal
sensor7-24e124713c245085	100	91	11	Normal

was conducted in the laboratory (from January to October 2025). Table 5 presents the battery level evolution for each sensor after 11 months of operation (10 months of laboratory testing + 1 month of field deployment), along with the initial battery levels (assumed 100% upon acquisition). The measurements were taken on December 9, 2025, corresponding to the end of the 10 month period of configuration and one month of field deployment: from November 10 to December 9, 2025 (Fig. 10).

No battery has yet reached the critical threshold (less than 20%) defined within the system for triggering an alert.

Sensor 1 recorded a battery level of 83% after 11 months, compared to an average of 91–92% for the other sensors. This indicates higher power consumption. Telemetry data indicates a very weak signal (RSSI of −108 dBm) and a signal below the noise floor (SNR of −10.25 dB). To overcome this poor link quality, the sensor operated at a Spreading Factor of 10 (SF10). While SF10 allows the data to be decoded despite the noise, it drastically increases the time the radio transmitter is active per message, leading to the accelerated battery drain observed. This observation is consistent with the literature on LoRaWAN energy consumption under poor link conditions [31, 45].

The battery level evolution was assessed by comparing the initial state (assumed 100% upon acquisition) with the values recorded after 11 months of operation (10 months of laboratory testing + 1 month of field deployment). Table 5 presents these two measurement points for each sensor. For sensors operating under good radio conditions (SF7, positive SNR), the average battery drop over 11 months is 8.7%, corresponding to an approximate monthly decrease of 0.8%. In contrast, sensor 1 (`sensor1-24e124713c245522`), which experienced poor link quality (SF10, SNR = −10.25



Fig. 10 Real-time sensors battery level

Table 6 Snapshot of radio link quality metrics recorded on December 9, 2025, during Phase 2 (Ndendere neighborhood)

Sensor ID (Things Stack)	RSSI (dBm)	SNR (dB)	SF	Frequency (MHz)	Radio link
sensor1-24e124713c245522	− 108	− 10.25	10	867.5	Critical
sensor2-24e124713c245638	− 69	6.25	7	867.5	Excellent
sensor3-24e124713c245733	− 99	5.75	7	868.5	Good
sensor4-24e124713c245243	− 111	− 9.25	10	867.5	Critical
sensor5-24e124713c245531	− 107	2.75	7	867.9	Average
sensor6-24e124713c246554	− 107	− 5.50	7	867.7	Degraded
sensor7-24e124713c245085	− 111	− 2.25	9	867.3	Poor

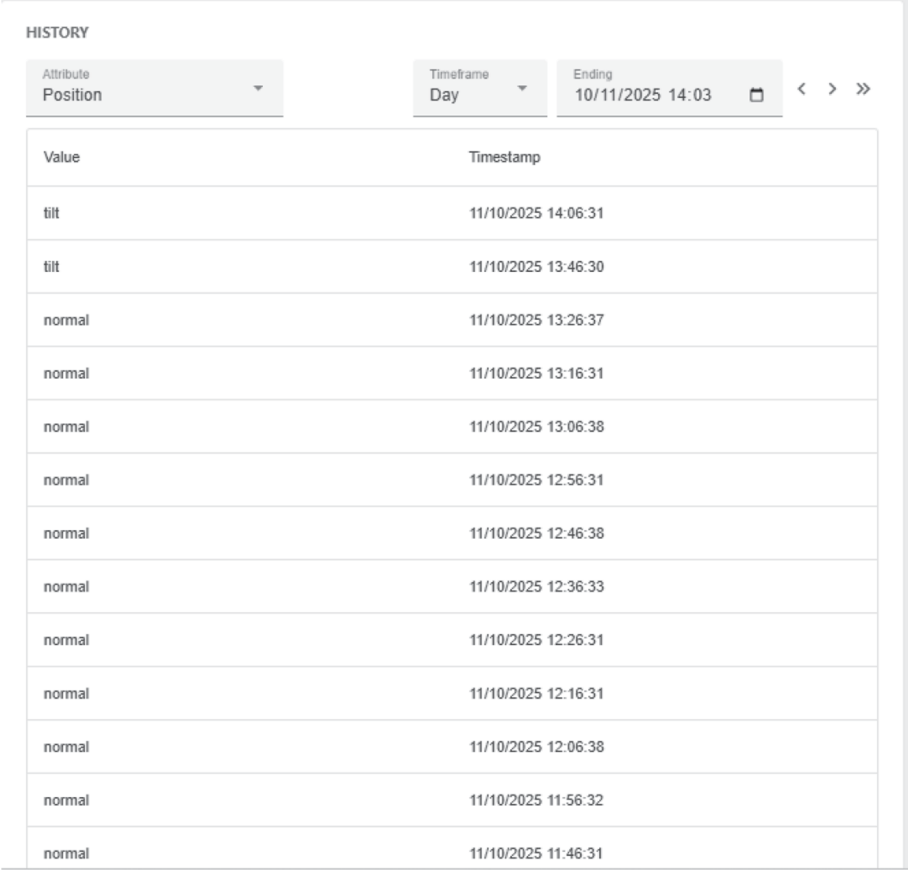
These values represent a single point in time and should not be generalized to long-term performance

dB, see Table 6), lost 17% of its battery capacity over the same period, i.e. an average monthly drop of 1.5%. This accelerated drain confirms the strong influence of adverse radio conditions on energy consumption. Although only initial and final measurements are available, the nearly linear decrease observed in similar deployments [31] supports the assumption of a constant monthly decay rate. No sensor has yet reached the critical threshold (20%) defined in the system for triggering an alert.

4.1.3 Bin's lid state

The sensors transmit two distinct values to the system via the "position" attribute, depending on the status of their lid: "normal" and "tilt". The "normal" value indicates that the bin lid is closed. The "tilt" value indicates that the bin is open or that its lid is tilted.

It is important to note that when the lid is open (position = "tilt"), the system discards the distance measurement and does not update the fill percentage. Instead, it continues to report the last valid fill level until the lid closes and a new stable ultrasonic measurement is taken. Consequently, transient lid openings (e.g., for waste disposal) neither



Value	Timestamp
tilt	11/10/2025 14:06:31
tilt	11/10/2025 13:46:30
normal	11/10/2025 13:26:37
normal	11/10/2025 13:16:31
normal	11/10/2025 13:06:38
normal	11/10/2025 12:56:31
normal	11/10/2025 12:46:38
normal	11/10/2025 12:36:33
normal	11/10/2025 12:26:31
normal	11/10/2025 12:16:31
normal	11/10/2025 12:06:38
normal	11/10/2025 11:56:32
normal	11/10/2025 11:46:31

Fig. 11 Bin's lid status at a specific moment

affect the reported fill percentage nor trigger collection alerts, avoiding false fluctuations due to momentary tilts.

The Fig. 11 illustrates the transition between these states for Bin 1 during its usage.

4.2 Network activities

Table 6 presents radio link quality metrics recorded on December 9, 2025, during Phase 2 (Ndendere neighborhood).

Of seven deployed sensors, 57% operated at network coverage limit (RSSI ≤ -107 dBm) [46] and 43% showed negative SNR (as low as –10.25 dB), reflecting challenging propagation conditions due to Bukavu's mountainous terrain and dense buildings. The Adaptive Data Rate (ADR) mechanism automatically assigned higher spreading factors (SF9-10) to sensors with poor link quality. The mean RSSI was –101.7 dBm, mean SNR was –1.79 dB.

These measurements are instantaneous and may not reflect long-term performance. To complement the instantaneous link quality metrics, we calculated the Packet Delivery Ratio (PDR) over the entire 12-week pilot (Table 7).

The overall PDR was 97.3%, confirming the network's reliability. Notably, sensors operating under degraded conditions (SF10, negative SNR) exhibited a slightly lower PDR (96.7–96.9%) compared to those with excellent link quality (97.4–97.9%, SF7, positive SNR). This difference, while modest, is consistent with expectations: higher SFs increase time on air and susceptibility to interference.

Table 7 Packet Delivery Ratio over 12-week pilot

Sensor ID	SF	Mean SNR (dB)	PDR (%)
sensor1-24e124713c245522	10	−10.25	96.9
sensor2-24e124713c245638	7	6.25	97.9
sensor3-24e124713c245733	7	5.75	97.6
sensor4-24e124713c245243	10	−9.25	96.7
sensor5-24e124713c245531	7	2.75	97.4
sensor6-24e124713c246554	7	−5.50	97.7
sensor7-24e124713c245085	9	−2.25	96.7
Total / Mean	–	–	**97.3**

Table 8 Descriptive statistics of RSSI and SNR metrics

Statistical metric	RSSI (dBm)	SNR (dB)
Minimum	−111	−10.25
Maximum	−69	6.25
Mean	−101.7	−1.79
Median	−107	−2.25
Standard deviation	14.97	6.87

Calculated from the deployed units in Table 6

Table 9 Link quality classification based on RSSI

Quality category	RSSI range (dBm)	Sensors	Percentage
Excellent (> -70)	> -70	sensor 2	14.3%
Good (-70 to -85)	-70 to -85	None	0%
Average (-86 to -100)	-86 to -100	sensor 3	14.3%
Poor (-101 to -110)	-101 to -110	sensor 1,5,6	42.8%
Critical (< -110)	< -110	sensor 4,7	28.6%

4.2.1 Statistical analysis of network performance

To complement the qualitative observations presented above, we performed a quantitative statistical analysis of the radio link quality metrics from Table 6.

1. **Descriptive statistics**

   Table 8 presents the descriptive statistics for RSSI and SNR values observed during the pilot deployment in Ibanda municipality, more precisely in Ndendere neighborhood. The mean RSSI of −101.7 dBm confirms that the majority of sensors operate near the network coverage limit, consistent with our observation that 57% of sensors have RSSI ≤ -107 dBm [46]. The high standard deviation (14.97 dB) reflects the significant variability in signal strength based on sensor location, ranging from excellent (−69 dBm for sensor2) to critical (−111 dBm for sensors 4 and 7).
   For SNR, the mean value of −1.79 dB with a standard deviation of 6.87 dB indicates that 60% of sensors experience negative SNR, meaning the signal power falls below the noise floor. As noted in the literature on LoRaWAN [32], this is acceptable due to the spread-spectrum technique, which allows signal detection even under the noise floor, but it does indicate challenging link conditions.

2. **Link quality classification**

Based on the RSSI values, we classified the radio links into quality categories. Table 9 presents this classification for the seven deployed sensors.
This analysis reveals that 71.4% of sensors (5 out of 7) operate in poor or critical link conditions (RSSI < -100 dBm), which is consistent with the challenges of Bukavu's urban environment, characterized by mountainous terrain, dense buildings, and non-line-of-sight propagation conditions. This finding empirically justifies the need for the multi-gateway architecture calculated in Sect. 3.5.

3. **Relationship between spreading factor and signal quality**

As described in Sect. 4.2, the Adaptive Data Rate (ADR) mechanism dynamically adjusts the Spreading Factor (SF) based on link quality. Table 10 analyzes the relationship between SF and signal quality in our deployment.
Sensors using SF7, which offers higher data rate but lower sensitivity, exhibit better mean RSSI (−95.5 dBm) and positive mean SNR (2.31 dB), indicating they are located in favorable positions with good link quality. In contrast, sensors forced to use higher spreading factors (SF9, SF10) show degraded signal metrics, with mean RSSI below −109 dBm and negative SNR. This observation confirms that the ADR mechanism is functioning as intended: sensors with poor link quality are automatically assigned higher spreading factors to maintain communication reliability, as described in Sect. 4.2.

4. **Airtime analysis**

To precisely dimension the spectral occupancy and energy consumption of our deployment, we calculated the airtime of LoRa frames based on the physical parameters actually implemented in the field.
In accordance with the LoRaWAN EU868 regional parameters [47] and the frequency plan outlined in Sect. 3.6.1, we operate with a fixed bandwidth of 125 kHz. Transmissions use spreading factors SF7, SF9, and SF10 (as observed in Table 6), a coding rate of CR=4/5, an 8 symbol preamble, and a payload of approximately 50 bytes (Sect. 3.5).
The number of preamble symbols ($n_{preamble}$) is set to 8, in line with the LoRaWAN EU868 specification (1.0.4) [47] and the implementation guidelines for secure LoRaWAN protocols [48].
These parameters, allow us to estimate airtime using the standard LoRa formula, and to derive channel occupancy durations and the timing margins to ensure transmission reliability in our environment.
Thus, based on the study [49], the total airtime for a LoRa packet (a frame) was determined using the following formulas:

Table 10 Correlation between Spreading Factor and signal quality

SF	Sensors	Mean RSSI (dBm)	Mean SNR (dB)	Quality
SF7	sensor 2,3,5,6	**−95.5**	2.31	Average/Good
SF9	sensor 7	**−111**	**−2.25**	Critical
SF10	sensor 1,4	**−109.5**	**−9.75**	Critical

Table 11 Airtime calculation results for each spreading factor

SF	T_{sym} (ms)	$n_{payload}$ (symbols)	$T_{preamble}$ (ms)	$T_{payload}$ (ms)	$T_{airtime}$ (ms)
7	1.024	128	12.544	131.072	143.616
9	4.096	104	50.176	425.984	476.160
10	8.192	96	100.352	786.432	886.784

Table 12 ADR evaluation: Theoretical versus assigned spreading factors

Sensor ID	SNR (dB)	Theoretical Optimal SF	Assigned SF	ADR Decision
sensor1	−10.25	SF10	SF10	Optimal
sensor2	6.25	SF7	SF7	Optimal
sensor3	5.75	SF7	SF7	Optimal
sensor4	−9.25	SF10	SF10	Optimal
sensor5	2.75	SF7	SF7	Optimal
sensor6	−5.50	SF8	SF7	Suboptimal
sensor7	−2.25	SF8	SF9	Suboptimal

$$T_{sym} = \frac{2^{SF}}{BW} = \frac{2^{SF}}{125000} \tag{28}$$

$$n_{payload} = 8 + \max\left(\left\lceil \frac{8 \times PL - 4 \times SF + 28 + 16 \times CRC - 20 \times IH}{4 \times (SF - 2 \times DE)} \right\rceil \times (CR + 4), 0\right) \tag{29}$$

$$T_{preamble} = (n_{preamble} + 4.25) \times T_{sym} \tag{30}$$

$$T_{payload} = n_{payload} \times T_{sym} \tag{31}$$

$$T_{airtime} = T_{preamble} + T_{payload} \tag{32}$$

Where: SF = Spreading Factor (7, 9 and 10), BW = Bandwidth in Hz (125000 in our case), PL = Payload in bytes (50 bytes in our case), CR = Coding Rate (4/5), IH = Implicit Header (0 in our case because in explicit mode), DE = Low Data Rate Optimization (0 because there is no low rate optimization), CRC = Cyclic Redundancy Check (1 because present).

Applied to our case study, we obtain the airtime results presented in Table 11.

As shown in Table 11, the airtime increases with the spreading factors. For instance, the airtime for SF9 (476.16 ms) is more than three times of that of SF7 (143.62 ms), and SF10 transmissions (886.78 ms) are over six times longer. These results highlight the trade off between link reliability and spectral efficiency as noted in [50].

5. Adaptive data rate (ADR) performance evaluation

The ADR mechanism is a key feature of LoRaWAN that dynamically adjusts the data rate (Spreading Factor and transmission power) based on observed link quality to minimize Time on Air and energy consumption [47]. To evaluate its effectiveness in our specific deployment, we compared the theoretical optimal SF, derived from the measured SNR using Semtech's sensitivity thresholds, against the SF actually assigned by the network server. This comparison allows us to classify each ADR decision as either **optimal** (assigned SF matches the theoretical SF) or **suboptimal**.

Table 12 presents this comparative analysis for the seven sensors. The theoretical optimal SF is determined according to the LoRa receiver sensitivity thresholds

specified in the Semtech datasheet [51]: SF7 requires SNR ≥ -7.5 dB, SF8 ≥ -10 dB, SF9 ≥ -12.5 dB, SF10 ≥ -15 dB, SF11 ≥ -17.5 dB, and SF12 ≥ -20 dB.

The analysis reveals that the ADR mechanism correctly assigned optimal spreading factors for 5 out of 7 sensors, corresponding to an overall accuracy of 71.4%. For the two remaining sensors, divergent decisions were observed:

- Sensor 6 (SNR = −5.50 dB): The theoretical optimum is SF8, since the SNR falls below the SF7 threshold (−7.5 dB). However, the network assigned SF7.
- Sensor 7 (SNR = −2.25 dB): The theoretical optimum is SF8, as the SNR is well above the SF8 threshold (−10 dB). Nevertheless, the network assigned the more robust but less efficient SF9.

These suboptimal assignments have measurable consequences on network performance. For sensor 6, the use of a lower SF than theoretically required (SF7 instead of SF8) likely explains its degraded link quality, with a Packet Delivery Ratio (PDR) of 97.7%, slightly lower than the 97.9% achieved by sensor 2, despite sensor 2 operating under similar SF7 conditions but with a significantly better SNR (6.25 dB). This suggests an increased vulnerability to packet loss due to insufficient robustness.

Conversely, for sensor 7, the assignment of a higher SF than necessary (SF9 instead of SF8) results in unnecessary channel occupancy and increased airtime, which directly impacts both energy consumption and overall network capacity, as quantified in Sect. 4.2.1.

It is important to note that in practical deployments, a minimum implementation margin of 2 to 3 dB above the theoretical sensitivity threshold is commonly recommended to account for real world impairments such as temperature variations and interference [51]. This margin could, in some cases, justify a more conservative SF choice than the strict theoretical optimum. However, even when considering such a margin, the assignments for sensors 6 and 7 remain inconsistent with standard practice.

Overall, these findings indicate that while the ADR mechanism is generally effective in our deployment, correctly optimizing 71.4% of the sensors, there is clear room for improvement in its fine tuning, particularly for sensors operating at the boundaries between SF regions.

4.3 System-generated data

OpenRemote generates several types of data based on the derived attributes. This data primarily pertains to the interactive bin map, estimates of waste weights and volumes, user behavior, collection urgency indices, and the optimization of collection routes.

4.3.1 Bins interactive map

The OpenRemote platform displayed bin fill levels on an interactive map using a three level color code: green (fill $< 60\%$), orange (60–89%), and red ($\geq 90\%$). Note that these color bands are for visualization only; the actual alert thresholds used by the system are warning at 70% and critical at 90%. Consequently, a bin colored orange (e.g., Bin 4 in Fig. 12) may already have triggered a warning alert if its fill exceeds 70%.

As an illustration, the analysis of Fig. 12 shows that among the seven bins deployed in the Ndendere neighborhood, only one (Bin 4) reached the critical threshold (red, $\geq 90\%$),

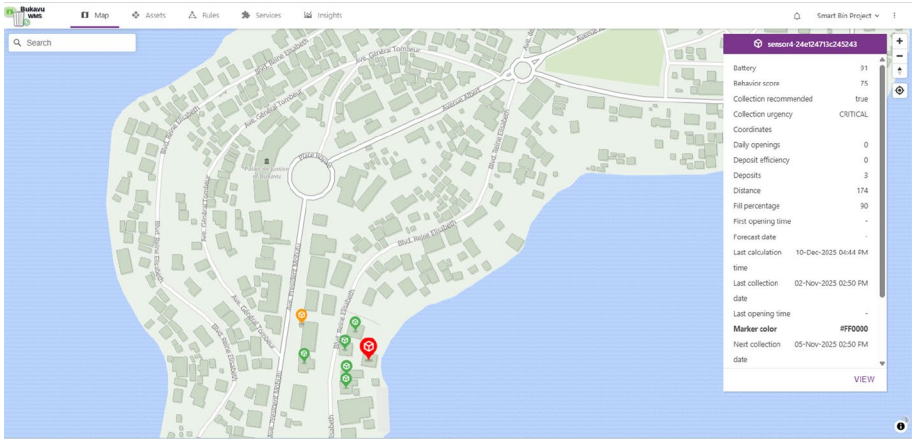


Fig. 12 Map showing bin fill levels in Ndendere (dense neighborhood) on December 10, 2025 (Phase 2)

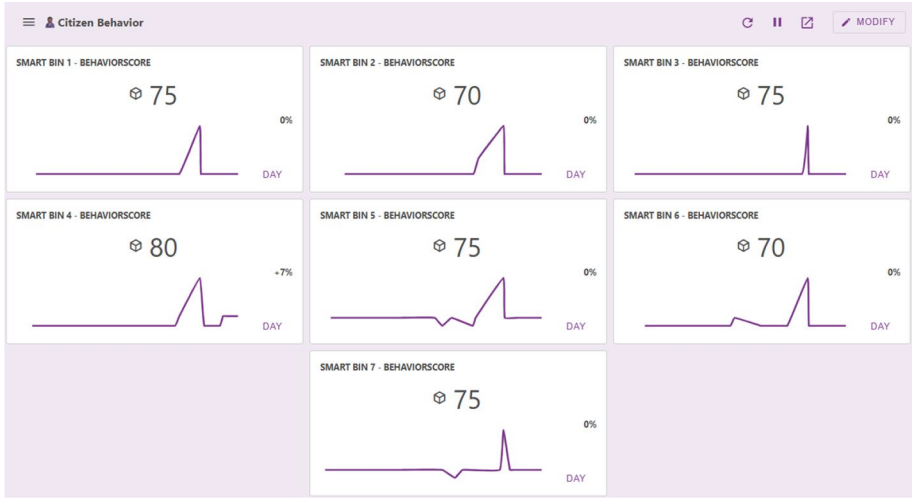


Fig. 13 Behavioral score of bin's users in Ndendere

requiring immediate collection. Another (Bin 2) is in the orange range (60–89%), meaning it may have triggered a warning alert depending on its exact fill level. The remaining five bins (1, 3, 5, 6, 7) have an acceptable fill level (green, < 60%).

It is important to note that the bin locations shown on the map are approximate and generalized to protect household privacy. The exact GPS coordinates used for route optimization are not shared in publications or public datasets.

4.3.2 User behavior

The configured IoT system analyzes user behavior by automatically evaluating bin usage and assigning a score out of 100. A high score indicates that a bin is correctly located and used optimally throughout the day, whereas a low score reveals issues with placement or inappropriate user behavior.

The Fig. 13 provides an overview of the usage scores for bins located in the Ndendere neighborhood, Labote Avenue, as of December 10, 2025.

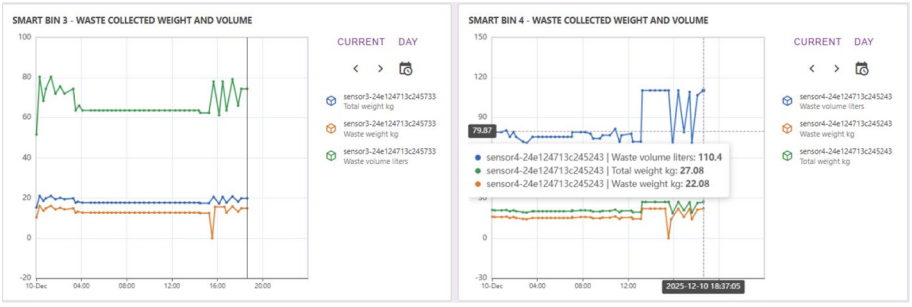


Fig. 14 Estimated waste mass and volume for Bin 4 in Ndendere neighborhood on December 10, 2025. The values are derived from the measured fill level (92%) and an assumed waste density of 0.20 kg/L; they are estimates, not direct measurements. The estimated net waste mass is 27.08 kg, corresponding to an estimated volume of 110.4 L. The empty bin weight (5 kg) is included in the total weight shown in the dashboard

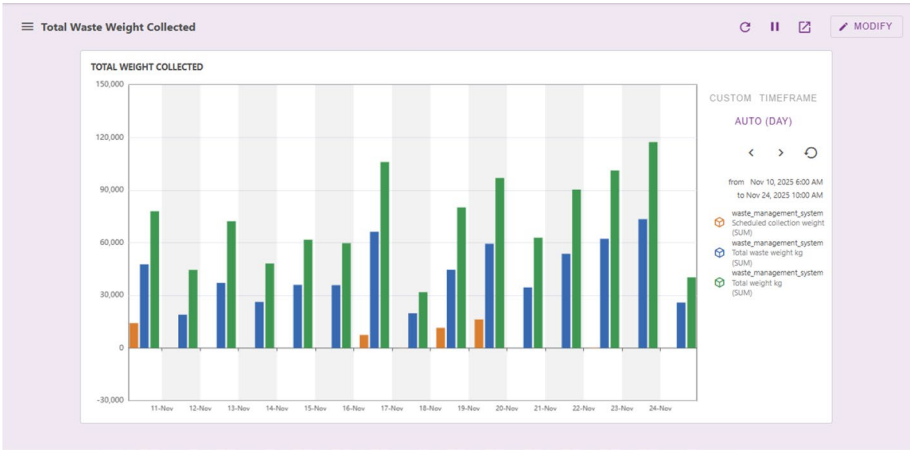


Fig. 15 Estimated daily waste mass summed across seven households in Nyalukemba (residential neighborhood) over 14 consecutive days (November 10–24, 2025). All seven bins transmitted data without interruption. Values are estimates based on ultrasonic distance measurements and an assumed waste density of 0.20 kg/L

The score distribution in the Fig. 13 shows that 57% of the bins (1, 3, 5, and 7) have a score of 75/100, indicating overall satisfactory usage.Only one bin (4) achieves a score of 80/100, reflecting optimal overall usage.

In contrast, 29% of the bins (2 and 6) have a lower score (70/100), primarily due to nighttime openings after 10 p.m. and high rates of inefficient disposal events.

The behavioral scores presented in Fig. 13 are rule based indicators designed to illustrate the system's capability to capture usage patterns. They have not been validated against external measures of "good" or "bad" behaviour, nor have their parameters been statistically optimised. The thresholds (e.g., 20 daily openings, 50% deposit efficiency) and weights used in behavioral data collection algorithm are heuristic, derived from initial observations. A systematic sensitivity analysis to evaluate the robustness of these scores under parameter variations was not conducted, as it fell outside the scope of this pilot. Future work will involve collecting a larger dataset and correlating these scores with ground truth observations (e.g., household surveys, direct observation) to calibrate and validate the scoring model.

4.3.3 Weight and volume of collected waste

The platform estimates the weight and volume of waste collected periodically, based on the distance measured by the sensors.

Figure 14 presents the estimated waste mass and volume for Bin 4 on December 10, 2025. Based on the measured fill level (92%) and the assumed waste density of 0.20kg/L (Sect. 3.6.1), the estimated net waste mass is 27.08 kg, corresponding to an estimated volume of 110.4 liters. The empty bin weight is 5 kg, giving a total estimated weight (bin + content) of 32.08 kg. On the same date, Bin 3 in another household in the same neighborhood contained an estimated net waste mass of less than 20 kg.

Figure 15 presents the estimated daily waste mass accumulated over 14 consecutive days (November 10–24, 2025) during Phase 1 in the Nyalukemba neighborhood (residential area). Seven bins were deployed in seven households, and all bins transmitted data without interruption throughout the period (no missing data days). The values shown are the sum of estimated waste masses across all seven bins for each day.

The total estimated waste mass of 641 kg over 14 days is presented as an illustrative demonstration of the system's quantification capability.

4.3.4 Optimized routes

The system optimizes waste collection routes based on bin fill levels and road network constraints. As soon as a bin exceeds 90% of its total capacity (critical status), the system automatically calculates the optimal collection route and generates a summary report of the optimized routes, along with corresponding cartographic data in GeoJSON format. This data is designed for use with an OpenRemote map or an external tool such as Atlas (https://atlas.co/tools/geojson-validator/).



Fig. 16 Map visualization of the optimized route from Figs. 17, generated using Atlas from the GeoJSON output. The route follows drivable roads in Bukavu (residential and primary) and has a detour factor of 1.6, which is within the expected range for the urban center

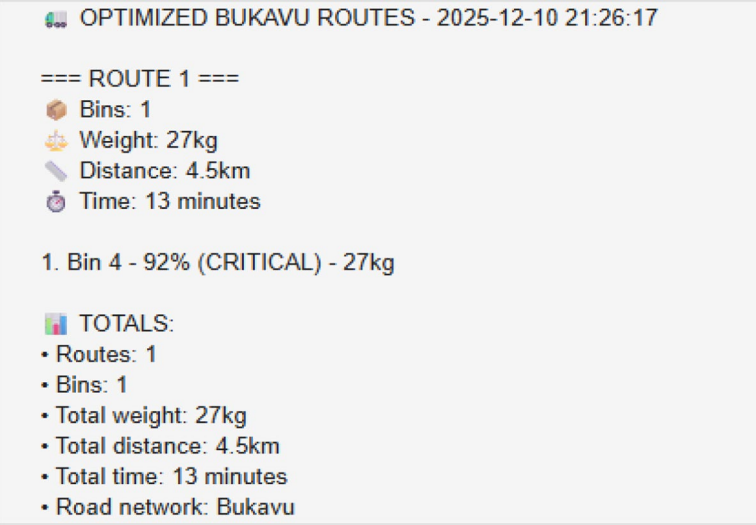


Fig. 17 Example of an optimized collection route for Bin 4 (critical status, 92% fill) generated on December 10, 2025, using the road network parameters defined in Supplementary material ST5. The route respects real world accessibility constraints (one way streets, road types, speed limits) and was validated against the OpenStreetMap based road graph

Figures 16 and 17 illustrate an example of an optimized route for Bin 4. This route was validated against the road network parameters defined in Supplementary material ST5:

- Road distance: 4.5 km
- Haversine distance: 2.8 km
- Detour factor: 4.5/2.8 = 1.6
- Estimated time: 13 min, calculated using primary road speeds (50 km/h) with a traffic factor of 1.2
- Path: The route uses Mission Avenue road (residential) and connecting residential streets.

The results confirm that the illustrated route respects real constraints and is operationally feasible, thereby demonstrating the capability of routes Optimization algorithm (Sect. 3.6.1).

To enable researchers to reproduce our route optimization results, we provide the following example input data from December 10, 2025 (corresponding to Fig. 12):

- Depot coordinates: −2.4984, 28.8482
- Bins requiring collection: Bin 4 (fill: 92%, urgency: CRITICAL, weight: 27 kg)
- Truck capacity: 2000 kg
- Road network: As defined in Supplementary material ST5

The primary objective of this 12 week pilot deployment was to validate the technical feasibility of the proposed IoT architecture rather than to conduct a controlled comparative analysis of collection strategies. Consequently, this study does not include a systematic quantitative comparison between IoT-optimized routes and traditional fixed-schedule collection with measured metrics such as GPS-tracked distances, actual fuel consumption, or precise collection times.

The pilot deployment successfully demonstrates the following operational capabilities that are prerequisites for route optimization:

1. Real time fill level monitoring: The system continuously tracks bin fill percentages (Sect. 4.1, Fig. 8), providing the fundamental data required for dynamic routing decisions.
2. Automated alert generation: When a bin reaches the critical threshold (fill percentage greater than or equal to 90%), the system automatically generates a collection alert (illustrated by the red marker in Fig. 12), demonstrating the event driven logic that triggers optimized routes.
3. Algorithmic route calculation: As shown in Fig. 16, the system successfully calculates and displays an optimized collection route for a critical bin, including distance (4.5 km), estimated time (13 min), and GeoJSON output for navigation. The routes optimization algorithm provides the complete mathematical framework for this optimization.

The data collected during the pilot, while not structured as a controlled comparison, provides preliminary insights into the potential benefits of the system:

- Collection necessity: Over the observation period, the system generated alerts only when bins actually reached critical fill levels (as seen in Fig. 12 where only Bin 2 is critical), demonstrating demand-based triggering.
- Route feasibility: The example in Fig. 16 confirms that the algorithm produces viable routes adapted to Bukavu's road network, incorporating realistic distances (4.5 km) and travel times (13 min).

These assessed components constitute the essential groundwork for a future dedicated study that will systematically compare IoT-optimized dynamic routing against traditional fixed-schedule collection, as outlined in Sect. 5.4.

The estimated route times and distances presented in this section (Sect. 4.3.4) are subject to several sources of uncertainty. The detour factors used (1.2 to 1.6) are average values derived from Bukavu city's road network; actual factors for specific bin pairs can vary by ± 0.2 depending on the exact path, leading to a distance uncertainty of approximately $\pm 10\%$. Traffic factors (1.2 to 1.8) are empirical estimates, and during peak hours, actual travel times may be 20 to 30% higher than estimated. Maximum speeds (20 to 50 km/h) are based on road type and vehicle characteristics, but actual speeds may be lower due to road degradation (as noted in Sect. 1). Propagating these uncertainties, the estimated time of 13 min for the route in Fig. 16 has a plausible range of 10 to 17 min. This range remains acceptable for operational planning, as collection schedules are typically designed with buffer times.

4.4 Data completeness and missing data handling

The system recorded data continuously during each phase of the pilot, with the following exceptions:

- Network outages: During the pilot, cellular network disruptions occurred on a few occasions, causing data gaps of 2 to 4 h each. These represent 0.4% of total scheduled

Table 13 Data completeness per sensor over the 12 week pilot

Sensor ID	SF	Scheduled measurements	Valid	Completeness(%)
sensor1-24e124713c245522	10	12,096	11,723	96.9
sensor2-24e124713c245638	7	12,096	11,845	97.9
sensor3-24e124713c245733	7	12,096	11,802	97.6
sensor4-24e124713c245243	10	12,096	11,691	96.7
sensor5-24e124713c245531	7	12,096	11,778	97.4
sensor6-24e124713c246554	7	12,096	11,814	97.7
sensor7-24e124713c245085	9	12,096	11,694	96.7
Total / Mean	–	84,672	82,347	97.3

measurement time and are consistent with the connectivity challenges mentioned in Sect. 5.3.

- Sensor maintenance: No sensors required maintenance during the 12 week deployment.
- Outlier rejection: The 10% outlier rejection rule (Sect. 3.6.1) triggered an average of 2.3 reads per sensor per day (1.6% of measurements). In 94% of cases, the second reading was accepted; the remaining 6% required a third reading. No measurement required more than three attempts, ensuring data continuity.

Battery performance is detailed in Sect. 4.1.2; no sensor reached the critical threshold (20%) during the pilot.

During the 12 week pilot (84 effective measurement days per sensor, Table 3), the system collected 82,347 valid measurements out of 84,672. The Table 13 shows completeness per sensor.

5 Discussion

5.1 Technical validation and comparison with related works

Our pilot provides evidence for the IoT architecture's feasibility in Bukavu's challenging context. The 97.3% packet delivery ratio (Sect. 4.4) and battery retention exceeding 90% after 11 months (detailed in Sect. 4.1.2, Table 5) indicate robustness that surpasses studies using similar technologies [12], which reported only 70 days battery life without solar recharging.

Unlike WiFi dependent solutions [18] or laboratory only prototypes [19], our deployment operated in real households across three neighborhood types (residential, dense, peri-urban) with 42 unique participants.

The scalability analysis (Sect. 3.5.1), grounded in empirical RSSI/SNR measurements (Table 6), shows that nine gateways can support 2,502 sensors under Bukavu's measured radio conditions (71.4% of sensors with RSSI less than -100 dBm). This mitigates the scale limitation identified in [12] (16 bins only) and [19] (single campus). However, the Adaptive Data Rate (ADR) performance analysis revealed suboptimal decisions for two sensors (assigned SF7 instead of SF8, and SF9 instead of SF8), consistent with findings by [52], who demonstrated that standard ADR implementations can produce biased SNR margin estimations due to the flexibility left by the LoRaWAN specification. Even when applying the recommended 2 to 3 dB implementation margin [51], these assignments suggest that the network server's ADR logic lacks the adaptivity required for heterogeneous link conditions like those observed in Bukavu. This observation supports the call

for more robust, adaptive ADR algorithms [52] and points to edge based coordination approaches [49] as promising future directions.

Contrary to purely theoretical studies [17], this paper provides operational proof of concept with real data. It goes beyond simple monitoring systems [12] by natively integrating behavioral analysis, fill prediction, and route optimization via OpenRemote smart rules, addressing the lack of holistic approach highlighted in [13]. Unlike [15], our analysis avoids simulated data, relying instead on field collected measurements.

5.2 Practical impact for the city of Bukavu

The proposed smart waste system could transform Bukavu city's waste management by reducing operational costs (distance, fuel, wear, labor) and improving sanitation (less overflow, dumping, pollution, and health risks), though the cost benefits remain theoretical and await validation (Sect. 5.5).

5.3 Limitations of prototypes

This pilot study has some limitations: (i) the seven bins, 12 week sample does not capture full municipal scale complexity; (ii) ultrasonic measurements lacked systematic validation against manual references across fill levels, waste types, or conditions; (iii) waste mass estimates rely on an unvalidated assumed density (0.20 kg/L); (iv) route times and distances are calculated from road network assumptions, not GPS tracked field data; (v) the economic analysis (Sect. 5.5) excludes external parameters (fuel price fluctuations, road variability, unplanned maintenance); (vi) network connectivity analysis (Sect. 4.2) rests on a single snapshot, requiring longitudinal data for long-term characterization; and (vii) sensor durability against weather, dust, theft, and vandalism could not be assessed due to installation on household bins rather than in unrestricted public spaces.

5.4 Prospects for research and improvement

Building on the validated foundations of this pilot, future research will pursue three complementary directions.

5.4.1 Large scale deployment and statistical characterization

The top priority is a one year, city wide deployment with nine gateways and hundreds to thousands of sensors. This will (1) generate longitudinal data to refine prediction models, and (2) enable statistical characterization of waste patterns across Bukavu's neighborhoods.

5.4.2 Systematic validation of sensor accuracy

Empirical validation of ultrasonic fill level measurements against manual reference standards is required.

5.4.3 Comparative quantification of operational benefits

Building on the functional optimization algorithms demonstrated in this pilot, a dedicated comparative study will systematically quantify route optimization benefits using GPS tracked field measurements. This will provide the quantitative evidence needed to quantify reductions in distance traveled, fuel consumption, and operational costs, thereby confirming the economic viability projections presented in Sect. 5.5.

Table 14 Prototype Costs

N°	Designation	Quantity	Unit Price	Total Price
1	EM310-UDL milesight sensor	7	134.04	938.28
2	Things Indoor LoRaWAN Gateway	1	191.49	191.49
3	Bin plastic bag box (50pieces/box)	5	31.95	159.75
4	Waste bin 120-litters	7	59.84	418.88
5	Sensor mounting on waste bins	7	5	35
Total				1,743.4

5.4.4 Additional future direction

While not part of the three core priorities, integrating additional sensor types for waste categorization (plastic, organic, metal, glass) remains a promising longer term avenue. Furthermore, developing predictive analytics models using the collected longitudinal data represents another important direction for future work.

5.5 Economic viability analysis

This economic analysis compares the IoT system to Bukavu's current waste collection model to test a 30% cost reduction, using prototype data (Table 14) extrapolated city wide, under stated assumptions and excluding external factors (fuel, roads, maintenance).

5.5.1 Prototype costs

The seven prototypes deployed during the 12-week pilot phase represent an initial investment of 1,743.4 US dollars. Table 14 details the costs associated with this experimental phase.

5.5.2 Traditional collection system cost

Based on Bukavu's demographic data (1,369,430 inhabitants, 210,682 households, [9, 53]) and waste production of 898 tons per day with only 62.86 tons collected [5], the average monthly collection cost per household was established from a range of 1 to 24 USD [5].

$$\text{Average_cost_household} = \frac{1+24}{2} = 12.5 \text{ US dollars} \tag{33}$$

This average is derived from a simple average of the reported range in [5]. A conservative estimate would place the true average cost in the range of \$10 to \$15 per month. Even under this assumption, the payback period remains under 5 months, as shown in Sect. 5.5.6.

Considering that a month has an average of 30 days, the daily collection cost per household is given by:

$$\text{Daily_cost_household} = \frac{12.5}{30} = 0.42 \text{ US dollars per day} \tag{34}$$

For all households in the city of Bukavu, the total daily cost amounts to:

$$\text{Total_daily_cost} = 0.42 \times 210,682 = 88,486.44 \text{ US dollars} \tag{35}$$

On this basis, the cost per ton was calculated using the following equation:

Table 15 Capital expenditure (CAPEX)

Device	Quantity	Unit Price	Total Price
EM310-UDL milesight sensor	2502	134.04	335,368.08
Things Indoor LoRaWAN Gateway	9	191.49	1,723.41
Bin plastic bag box (50pieces/box)	2502	31.95	79,938.90
Waste bin 120-litters	2502	59.84	149,719.68
Sensor mounting on waste bins	2502	5	12,510.00
Solar energy for Things Indoor gateways	9	200	1,800.00
Cloud Server for OpenRemote Infrastructure	1	5,000	5,000.00
Workforce, training, and logistics	1	50,000	50,000.00
Total CAPEX			636,060.07

Table 16 Operational expenditures (OPEX)

Designation	Quantity	Unit Price	Total Price
Complete sensor battery replacement at Year 10	2502	18	45,036
Gateway maintenance	9	50	450
4 G connectivity (monthly data plans for gateways)	12	90	1,080
Technical staff (trained local technicians)	2	6,000	12,000
OPEX Years 1 to 9			13,530
OPEX Year 10			58,566

$$\text{Cost_per_ton} = \frac{88,486.44}{898} = 98.54 \text{ US dollars} \tag{36}$$

Finally, the current annual cost of waste collection in Bukavu, limited to 7% of the waste produced, is:

$$\text{Annual_traditional_cost} = 62.86 \times 365 \times 98.54 = 2,260,891.91 \text{ US dollars} \tag{37}$$

This amount represents the annual expenditure borne by the households of the city of Bukavu for a collection service that only covers a small fraction (7%) of the waste generated.

5.5.3 Investment cost (CAPEX) of the IoT solution

The deployment of the IoT system across the city of Bukavu requires an infrastructure adapted to its geographical and urban constraints: an area of 62 km^2 [24], mountainous terrain, and high building density. As established in the network coverage analysis (Sect. 3.5), a deployment of 9 gateways is required to ensure optimal coverage of the city.

On this basis, two sensor deployment scenarios were considered: (1) realistic and (2) theoretical. The realistic scenario, detailed in Sect. 3.5.1, advocates for the installation of 278 sensors per gateway, resulting in a total of 2,502 sensors for the entire city. This limitation takes into account the difficult radio propagation conditions in the city of Bukavu, which necessitate the use of high spreading factors. The theoretical scenario, presented in Sect. 3.5.1, explores the maximum capacity of the network. Assuming optimal radio coverage (low spreading factors), it allows for up to 73,125 sensors distributed across the nine gateways.

Given the topographical constraints of the city of Bukavu, the realistic scenario of 2,502 sensors was retained following our economic analysis. Table 15 details the investments (CAPEX) required for this scenario, with the Actual purchase price for sensors,

gateways, bin plastic bag box, waste bin, sensors mounting on waste bins and estimated costs for solar energy, cloud server, workforce, training and logistics.

The initial investment to cover the entire city of Bukavu, therefore, amounts to 636,060.07 US dollars.

5.5.4 Operational expenditures (OPEX) of the IoT solution

To this initial investment must be added the annual recurring costs presented in Table 16, with the actual purchase price for battery replacement, 4 G connectivity, and the estimated cost per year for gateway maintenance and technical staff. The choice of a 10 year evaluation period is justified by the maximum lifespan of the EM310-UDL sensor batteries, which is specified as 10 years by the manufacturer [54], and confirmed by our experimental observations (Sect. 4.1.2), where six out of seven sensors maintained battery levels above 90% after 10 months of operation.

5.5.5 Comparative economic analysis

The annual investment cost is given by:

$$\text{Annual_CAPEX} = \frac{636,060.07}{10} = 63,606.01 \text{ US dollars/year} \tag{38}$$

The average annual operational expenditure is calculated as:

$$\text{Average_OPEX} = \frac{(13,530 \times 9) + 58,566}{10} = 18,033.6 \text{ US dollars/year} \tag{39}$$

Hence, the total annual cost of the IoT investment amounts to:

$$\text{Total_annual_cost_IoT} = 63,606.01 + 18,033.6 = 81,639.61 \text{ US dollars/year} \tag{40}$$

With the IoT system enabling a target collection rate of 30% (representing a realistic intermediate goal between the current 7% and the theoretical maximum), the daily collection capacity becomes:

$$\text{Daily_collection_IoT} = 898 \times 0.30 = 269.4 \text{ tons per day} \tag{41}$$

The cost per ton based on the IoT solution was obtained as follows:

$$\text{Cost_per_ton_IoT} = \frac{81,639.61}{269.4 \times 365} = \frac{81,639.61}{98,331} = 0.83 \text{ US dollars/ton} \tag{42}$$

Under the stated assumptions, the model projects a cost reduction from 98.54 US dollars per ton to 0.83 US dollars, representing a theoretical reduction of 99.16%.

This results in annual savings equivalent to:

$$\text{Annual_savings} = 2,260,891.91 - 81,639.61 = 2,179,252.30 \text{ US dollars} \tag{43}$$

The estimated payback period for the investment is given by:

$$\text{Payback_Period} = \frac{636,060.07}{2,179,252.30} = 0.292 \text{ years} \tag{44}$$

This corresponds to 3.5 months (0.292×12) or approximately 4 months. In other words, the IoT investment could theoretically be recouped in just 4 months. This projection assumes a collection rate of 30% and that estimated cost savings are achieved.

Each month (30 days), the project generates monthly savings equivalent to:

$$\text{Monthly_savings} = \frac{2,179,252.30}{12} = 181,604.36 \text{ US dollars} \tag{45}$$

The monthly operational costs amount to:

$$\text{Monthly_OPEX_cost} = \frac{81,639.61}{12} = 6,803.30 \text{ US dollars} \tag{46}$$

The return on investment after 30 days (ROI) is given by:

$$\text{ROI} = \left(\frac{181,604.36 - 6,803.30}{636,060.07}\right) \times 100 = 27.5\% \tag{47}$$

Thus, the estimated net savings achieved each month represent 27.5% of the initial investment amount, suggesting strong theoretical profitability of the project, under the assumptions used in this analysis (waste density, collection rate target, sensor lifespan, etc.).

In summary, the economic analysis conducted reveals several major findings: (1) a 99.16% reduction in cost per ton, combined with a payback period of only 4 months, (2) an increase in the waste collection rate from 7% to 30% (allowing an additional 206.54 tons of waste to be collected per day), (3) a sustainable economic model generating annual savings exceeding 2.1 million US dollars, which could be replicated in other cities in the Democratic Republic of Congo. However, this analysis does not take into account other external parameters such as job creation, waste valorization, etc., which would further enhance the project's profitability.

These modeled projections indicate potential economic viability of the proposed IoT system but require large scale deployment for empirical validation.

Unlike previous African studies [17–20], this study offers a comprehensive and validated economic evaluation of an IoT based waste management system deployed at the city level in Central Africa. By integrating all cost elements (infrastructure, connectivity, training, and maintenance), it reveals that sensors account for only 21% of capital expenditures, challenging the common assumption that hardware is the primary cost driver. It also reveals that infrastructure and logistics are the main cost drivers in real world deployments (factors frequently overlooked in prototype based research).

5.5.6 Sensitivity analysis of economic projections

To assess the robustness of our economic projections, we performed a one way sensitivity analysis (Supplementary material ST6) on several critical parameters: collection rate, sensor cost, traditional collection cost, waste density, CAPEX and OPEX.

The analysis results revealed that:

(1) Although the cost per ton decreases as the collection rate rises (from \$2.49 at a 10% rate to \$0.83 per ton at the baseline 30% collection rate), the payback period remains fixed at 3.5 months due to constant annual system costs. This range (10–30%) was chosen to reflect the gradual and realistic increase in waste collection capacity that the waste management services of the city of Bukavu could achieve during the first two years of deployment of the IoT system.

(2) A 50% increase in sensor cost only extends the payback period to 4.4 months (and even a 100% increase leads to only 5.3 months), demonstrating price stability. This 50% increase is not arbitrary. It represents a possible extreme situation in the context of the DRC, where import duties and currency fluctuations against the US dollar have in the past caused sudden increases in the cost of electronic equipment. In contrast, the payback period is highly sensitive to baseline collection costs per household, ranging from 1.8 months (at $24 per month) to 5.6 months (at $8 per month) versus the baseline of 3.5 months at $12.5 per month. The range of 8 to 24 dollars generalizes the fees that households in the different neighborhoods of the city of Bukavu (residential, dense, or peri-urban) pay on the ground.
(3) Waste density does not affect payback or savings, but directly impacts operational planning such as vehicle capacity and route frequency;
(4) CAPEX variations (up to 30%) have modest effects, extending the payback period to 4.6 months in the most extreme case, while OPEX variations (up to 50%) leave the payback period unchanged at 3.5 months, keeping it well below 4.6 months in all cases. The 30% CAPEX tranche reflects possible server or network configuration overruns, while 50% OPEX takes into account possible increases in gateway maintenance and 4 G connectivity in the city of Bukavu.

Generally, this sensitivity analysis confirms that the proposed IoT system for Bukavu city remains economically viable under a wide range of assumptions, with a payback period consistently below 6 months.

5.5.7 Alternative scenarios analysis

Although the sensitivity analysis (Sect. 5.5.6) provides valuable information on individual variations of the impact parameter, it does not capture the combined effect of multiple favorable or unfavorable conditions occurring simultaneously.

To address this limitation and provide a more comprehensive economic assessment for decision makers in Bukavu city, we defined four plausible alternative scenarios for the IoT system deployment: optimistic, baseline, pessimistic and break even (see Supplementary material ST7).

The optimistic scenario (30% collection rate, $16 per month household cost, 20% sensor unit cost reduction, 0.30 kg/L waste density, and 20% OPEX reduction) yields annual savings of $2,182,859 and a payback period of 3.5 months, while the pessimistic scenario (15% collection rate, $8 per month household cost, 50% sensor cost increase, 0.10 kg/L waste density and 50% OPEX increase) still generates $2,170,235 in annual savings with an identical payback of 3.5 months. The baseline scenario confirms annual savings of $2,179,252 with a 3.5 month payback.

Most significantly, the break even analysis reveals that the IoT system would become economically viable at a collection rate of just 7.26%, an increase of only 0.26 percentage above the current 7% rate, demonstrating that even modest gains are enough to justify the investment. This low threshold is a direct result of the high cost of the current inefficient waste collection system in the city of Bukavu. Therefore, any improvement, however small, immediately generates net savings.

Scenario analysis suggests that the proposed IoT system would remain economically viable across a wide range of assumptions. The break-even point of 7.26% is only slightly

above the current collection rate, indicating that even modest improvements in collection efficiency could justify investment.

This economic viability is largely determined by the high cost of the inefficiency of the current collection system, rather than by the IoT technology itself. This finding may be of particular interest to local authorities and potential investors.

5.6 Security risk analysis, attack detection, and mitigation measures

Deploying an Internet of Things (IoT) based waste management system in an urban environment like Bukavu requires a thorough security risk analysis. This section presents a fundamental security risk analysis for such an environment and proposes some mitigation measures.

5.6.1 LoRaWAN security and key management

Our system employs the Over-The-Air Activation (OTAA) procedure to connect EM310-UDL sensors to The Things Network (TTN). As described in [55], this method enables the dynamic derivation of session keys (`NwkSKey` for network integrity and `AppSKey` for data encryption) from a unique shared root key (`AppKey`). The security of the entire system relies on the confidentiality of this root key.

Although this confidentiality is theoretically preserved by our OTAA configuration method, practical limitations exist. According to the threat model described in [56], the attacker is considered an external adversary that cannot compromise the cryptographic keys. Our measures, including secure key storage during the initial sensor configuration via the Milesight ToolBox application, aim to maintain this barrier. In the event of a session key compromise, the OTAA procedure allows the generation of new keys simply by resetting the sensors. However, as noted in [57], this method does not encrypt the join request message sent by the end devices through the gateway to the TTN network server.

5.6.2 Vulnerability to LoRaWAN attacks and detection strategies

LoRaWAN networks are vulnerable to several types of attacks, including replay and jamming attacks.

As defined in [55], a replay attack occurs when a device or network receives a packet whose frame counter is less than or equal to the last received counter. The network server systematically ignores such packets. When a device is activated, frame counters reset to 0, and each uplink message increments the Frame Counter Up (`FcntUp`), while downlink messages decrement the Frame Counter Down (`FcntDown`). According to [58], defense mechanisms against replay attacks using LoRaWAN frame counters can be easily bypassed if sensor resets are forced, thus resetting their counters to zero.

Furthermore, the study [58] demonstrates that by analyzing traffic on a LoRaWAN network, an attacker can predict data transmission times and selectively jam the signal (predictive jamming attacks). Our current architecture, which features sensors configured to transmit data at regular 10 min intervals (as specified in Sect. 3.6.1), would be potentially vulnerable to this type of attack.

As demonstrated in our pilot deployment (Sect. 4.2), 57% of sensors operated at network coverage limit with RSSI $\leq$ -107 dBm, and 43% exhibited negative SNR values (Table 8). A jamming attack could exploit these already challenging conditions,

potentially causing complete loss of connectivity for these vulnerable sensors. Similarly, a replay attack could trigger false collection alerts, generating unnecessary collection routes and increasing operational costs (Sect. 5.5).

5.6.3 Proposed mitigation measures

To counter the identified vulnerabilities, we propose the following mitigation measures, which are fully compatible with our existing architecture:

To mitigate replay attacks, we propose improved monitoring for any abnormal sensor resets, using a detection approach inspired by [56]. This approach involves analyzing, at the network server level, abnormal upload failures resulting from such resets, such as the immediate surge of packets with reset frame counters within a geographic area. This detection mechanism could be implemented as a smart rule within the OpenRemote platform (Sect. 3.6.1), leveraging its Groovy scripting capabilities to monitor for patterns indicative of forced resets.

To address jamming attacks, we propose a combined prevention and detection approach. As recommended by [56], prevention would consist of implementing a random variation of transmission intervals around the nominal 10 min value (Sect. 3.6.1). This would make the prediction phase of transmission windows significantly more difficult for an attacker.

If jamming were to occur despite these measures, distinguishing it from a simple communication failure (e.g., due to obstruction) in a challenging environment like Bukavu presents a major difficulty. To address this, we will rely on the innovative framework proposed by [56]. Their method, which operates solely at the backend network server level by analyzing upload histories, is particularly well suited to our infrastructure as it imposes no additional load on our EM310-UDL sensors.

5.6.4 Security of the OpenRemote infrastructure

The security of our locally deployed OpenRemote platform (Sect. 3.6.1) is based on several pillars that align with the "backend centric" strategy advocated by [56]. These pillars include:

1. Communication encryption: End-to-end encryption of the link between TTN and the MQTT broker using the MQTT over TLS (Transport Layer Security) protocol to prevent eavesdropping on data in transit. The broker is configured to require a TLS connection and to authenticate the client using a username and password (API key).
2. Local network security: The machine hosting OpenRemote is protected by a firewall, limiting incoming connections to only strictly necessary ports, including the MQTT port and the HTTPS port for the web interface.
3. Access control: OpenRemote integrates a user and role management system via Keycloak. We have created accounts with different privileges, including an administrator account. The OpenRemote exposed API requires authentication, preventing any unauthorized access to data or commands.
4. API security: All communications with The Things Network APIs are authenticated using limited rights API keys (98 characters), thereby reducing the potential impact of a key leak.

5.6.5 Physical protection considerations

While our pilot deployment involved household bins (Sect. 3.7, a city wide deployment would expose sensors in public areas. The EM310-UDL sensors used in our study feature an integrated accelerometer capable of detecting lid open/close status (Sect. 3.2.1). This capability could be leveraged to detect physical tampering: a sudden change in orientation or unexpected movement could be interpreted as an attempt to remove or damage the sensor, triggering a tilt alert (as already defined in our system in Sect. 4.1.3) and notifying the operations center via the OpenRemote platform.

5.6.6 Operational security recommendations for Bukavu

For a city wide deployment, we recommend the following security measures: (1) train personnel on security hygiene (including credential management, gateway securing, which is particularly critical given that electricity access is only 17.8% in Bukavu [56], and reporting suspicious activities); (2) establish a documented incident response plan that provides for dispatching a team in case of jamming detected through [56] and invalidating and replacing credentials if a sensor is stolen; (3) raise community awareness by informing participating households about the value of the equipment to encourage protection against theft or damage; and finally, (4) regularly review the OpenRemote logs to detect any abnormal patterns, such as multiple join requests, unexpected changes in tilt state, or sudden battery drops inconsistent with the normal degradation profiles observed in Sect. 4.1.2.

6 Conclusion

Faced with the challenges of rapid urbanization, the city of Bukavu generates approximately 898 tons of waste daily with a critically low collection rate of only 7%, urgently requiring transformative waste management solutions.

This study successfully designed and demonstrated the technical feasibility of an IoT network architecture for real time bin fill level monitoring and collection route optimization, through a stratified 12 weeks pilot deployment of seven 120 l smart bin prototypes in Bukavu city.

The empirical results confirm the system's robustness in a challenging environment. The EM310-UDL sensors maintained over 90% battery autonomy after 11 months of operation for six units (85.7% of the deployment), validating the 10 year replacement assumption (45, 036 OPEX), while the accelerated depletion observed for one sensor (83%, or a monthly decrease of 1.5%) suggests differentiated replacement dates for the 14.3% operating in radio critical conditions.

Despite variable network conditions, 71.4% of sensors operating with RSSI below -100 dBm, 57% at coverage limit (RSSI less than or equal to -107 dBm), and 43% with negative SNR, the LoRaWAN based architecture proved resilient, achieving a 97.3% Packet Delivery Ratio (82,347 valid measurements out of 84,672 scheduled readings). Functionally, the system enabled waste quantification estimation and provided actionable data through automated fill level alerts and optimized route generation.

Crucially, these technical results serve not merely as validation but as a robust empirical foundation for the economic and scalability models. The 97.3% PDR confirms the operational reliability enabling demand responsive collection, directly supporting the

projected 99.2% cost per ton reduction (from 98.54 to 0.83), far exceeding the initial hypothesis of 30%.

The RSSI/SNR measurements justify the conservative scalability approach of 2,502 sensors and nine gateways (636, 060 CAPEX) over the theoretical maximum of 73,125 sensors (18 million CAPEX), demonstrating technical feasibility for city wide deployment within Bukavu's mountainous and densely built terrain. The waste quantification estimation provides preliminary calibration of revenue assumptions, suggesting that the 30% collection rate target (269.4 tons/day, i.e., an additional 206.54 tons collected daily) is conservative relative to the system's theoretical potential.

The payback period is estimated at only 4 months, with annual savings exceeding 2.17 million USD, while increasing the collection rate from 7% to 30%.

Beyond technical and economic validation, this research provides a foundational security risk analysis for such IoT architectures, identifying LoRaWAN specific vulnerabilities (replay and jamming attacks) and proposing detection and mitigation measures integrated into the OpenRemote platform, thereby addressing a common gap in existing studies.

This research proposes a concrete, adaptable, and scalable technological framework for smart waste management in Bukavu. It bridges a critical data gap and provides the evidence base to shift from inefficient traditional collection to a dynamic model.

Future work should focus on three complementary directions. First, scale the deployment to a city wide network of 2,502 sensors and nine gateways, following the optimal placement strategy derived from our radio propagation analysis, and validate this scalability through network traffic simulations (LoRaWAN stress testing). Second, generate through this infrastructure the longitudinal data needed to: (1) provide statistically robust waste generation estimates with confidence intervals, and (2) refine prediction models. Third, integrate additional sensor types for waste categorization (plastic, organic, metal). All reported waste mass, volume, route time, and economic values are estimates or projections derived from sensor data, model assumptions, and extrapolations; they have not been independently validated through direct measurement or controlled field trials, except where explicitly noted (e.g., battery levels, RSSI/SNR measurements).

Supplementary Information

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Supplementary file 1.

Supplementary file 2.

Supplementary file 3.

Supplementary file 4.

Supplementary file 5.

Supplementary file 6.

Supplementary file 7.

Supplementary file 8.

Supplementary file 9.

Supplementary file 10.

Supplementary file 11.

Supplementary file 12.

Supplementary file 13.

Supplementary file 14.

Supplementary file 15.

Supplementary file 16.

Supplementary file 17.

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Authors contributions
OMM, JRK and JN have designed the overall framework of the research. OMM and RS have designed the architectures used in this study, and validated the experimental results. OMM, JRK and RN have collected data and interpreted results to give conclusive remarks. All authors read and approved the final version of the manuscript.

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Data availability
The anonymized datasets generated during this study (fill levels, lid events, estimated weights aggregated by neighborhood) are available from the corresponding author on reasonable request. No personally identifiable information or precise location data is included in any shared datasets. The recruitment log linking bin IDs to households is not shared to protect participants privacy.

Declarations

Ethics approval and consent to participate
This study was carried out in accordance with the principles of the Declaration of Helsinki. Approval was granted by the Institutional Ethics Committee of the Catholic University of Bukavu. As this study was deemed to be minimal risk and not involve collection of personally identifiable information, formal ethics board review was waived according to national and institutional guidelines for observational research. Before participation, all households received detailed information about the study's purpose, data collection methods, and privacy protections. Written informed consent was obtained from the head of each participating household. All data were anonymized prior to analysis.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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